



Trace and minor elements in the ore minerals: An assessment of geochemical distribution in overprinting epithermal on porphyry mineralization at Chodarchay, NW Iran

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ABSTRACT

The Chodarchay overprinting epithermal on porphyry mineralization occurs in the Tarom-Hashtjin subzone of the Alborz magmatic arc in northwestern Iran. Rock units include intrusions (phases I and II) and volcano-volcaniclastic rocks. The alteration types consist of potassic, phyllic, argillic, and propylitic. The porphyry type is mostly accompanied by potassic alteration. The Chodarchay Fault implies two extensional and compressional mechanisms. The porphyry type is associated with the extensional and the epithermal type with the compressional stage of the fault. Mineralization occurred mainly inside the Eocene volcanic and volcano-sedimentary rocks; some ores are associated with the Late Eocene quartz monzonite. The mineralization is divided into three zones from depth to surface: 1) Hypogene sulfide zone (deep); 2) Enriched supergene sulfide zone (near-surface); 3) Oxidized and leached zone (surface). The major sulfide minerals consist of chalcopyrite, pyrite, sphalerite, and galena. Based on the results of the new analyses, chalcopyrite hosts Ag as a solid-solution and contains Ag-rich Bi-sulfosalt micro-inclusions. Non-constant Cd/Zn ratio in the chalcopyrite can indicate varying physiochemical conditions during its crystallization. Low As contents in the pyrite caused relatively clean pyrite with very few elements. The Ni concentrations in the pyrite imply that it was sourced from mantle mafic-ultramafic rocks. Sphalerite is the only Au-host mineral among the sulfide minerals as a nano-mineral phase. Solid-solution Ag exists in the sphalerite. The sphalerite Fe/Zn ratios show that it was precipitated between 220 and 280 °C. Silver and Bi have the highest measured concentrations among all analyzed elements in the galena, mainly partitioning within the galena, but Hg and Cd are abundant in the sphalerite. Bornite is one of the major carriers of Ag and Bi. Silver and Bi exhibit considerable variations in the Cu-(Fe) sulfides. A new occurrence of sulfosalt minerals was identified at Chodarchay (N = 14). The discovered Bi-sulfosalts include cuprobismuthite group minerals (cuprobismuthite, hodrushite, and paderaitite), aikinite-bismuthinite series (aikinite and friedrichite), emplectite, makovickyite, galenobismutite, gustavite, heyrovskyite, neyite, cupropavanite; and As-Sb-bearing Cu sulfosalts (watanabeite and colusite). This number of sulfosalts had not previously been reported from any of the Tarom or Alborz deposits or mines. The Au-Ag occurrence was reported from the neyite, emplectite, and gustavite. The Ag-rich minerals crystallize from fractionated fluids that cooled during rising. Silver-carrier sulfosalts may have been exsolution from pre-existing Ag-bearing chalcopyrite in the cooling system. Most of the Ag-bearing sulfosalt minerals formed at the margins and fractures of the chalcopyrite, indicating that the cooling started from these points. Thus, Ag at Chodarchay is hosted inside Cu-(Fe) sulfides, the significant part of them being chalcopyrite and Ag-rich sulfosalt micro-inclusions inside the chalcopyrite. The fluid temperature for the Ag and chalcopyrite precipitation is from 200 to 350 °C. Bismuth assemblages precipitated at > 350 °C. The cuprobismuthite group crystallizes at temperatures > 300 °C. The supergene process and rare secondary enrichment of Ag and Au took place in the oxidation zone; this enrichment is dependent on the presence of manganese. Based on the fire-assay results, the concentration of gold is greater at shallower depths.

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1. Introduction

The Chodarchay deposit is located in the Tarom-Hashtjin (Tarom) subzone. During the past decades, many studies have been carried out on this subzone. In the following, the cases that have been published recently are mentioned such as the study of controls on epithermal and porphyry mineralization by the geochemistry of igneous rocks (Ghasemi Siani et al., 2020), investigating the possibility of association of porphyry type with epithermal type in Tarom subzone using fluid inclusions (Mousavi Motlagh et al., 2021), determining magma fertility by mineral chemistry in the Tarom intrusions (Yasami et al., 2022), the study of alteration types related to the precious and base metal epithermal deposits (Ghasemi Siani and Lentz, 2022), investigating the association between iron and copper mineralization and faults by multifractal modeling in Tarom (Adib et al., 2022), evolution of the tectono-magmatic and metallogenic considerations in the Tarom (Afshari et al., 2023), as well as geochemistry and petrogenesis of Tarom granitoid rocks (Ghasemi Siani and Ebrahimi Fard, 2023).

Many epithermal deposits were investigated and explored in this subzone including Abbasabad (Moradi et al., 2010; Jahandideh Kazempour et al., 2011; Moradi, 2011), Barik-Ab (Bazargani-Guilani and Parchekani, 2011), Khalifelou (Esmaeli et al., 2015; Hosseinzadeh et al., 2016), Nikoueieh (Aghajani Marsa et al., 2016), Goloujeh (Glojeh) (Mehrabi et al., 2016) Aghkand (Kouhestani et al., 2017a), Aliabad-Khanchay (Kouhestani et al., 2018), Shah-Ali-Beiglou (Mikaeili et al., 2018), Changoreh (Siahcheshm et al., 2018), Chargar (Mousavi Motlagh and Ghaderi, 2019; Mousavi Motlagh et al., 2019a,b), Marshoun (Kouhestani et al., 2019b), Lubin-Zardeg (Rahmani et al., 2019; Zamanian et al., 2019), Zajkan (Kouhestani et al., 2019a; Mahzi et al., 2022), Zehabad (Shahbazi et al., 2019, 2021), Rashtabad (Ajalli et al., 2021), Varmazyar (Ghorbani et al., 2022), Chomalou (Nazarian et al., 2023) and Jizvan (Golshani Nasab et al., 2023). Therefore, this subzone is an important host of epithermal mineralization in Iran, and exploration work on this type of deposits continues.

The chemistry of sulfide minerals is used for the exploration and mineral processing of porphyry and epithermal deposits. Understanding trace and minor elements hosted by sulfide minerals helps to appraise the partitioning and distribution of the precious constituents (like Au and Ag and also critical elements like Bi, As, and Sb; European Commission, 2023) among coexisting minerals. Such knowledge is used to optimize the processing and recovery of valuable by-products (for example, Ag within sulfides) during Cu smelting and refining (Robert et al., 2003). The concentrations and distribution of minor and trace elements within sulfides conduce mine-scale exploration.

Evaluating minor/trace element (not only Au and Ag but also other strategic and critical metals) distribution in any sulfide and sulfosalt mineral enables distinguishing among the elements present in a solid-solution and the micro-scale inclusion of other minerals. Electron Probe Micro Analysis (EPMA) trace-element data are presented for base metal sulfides as well as for accompanying sulfosalts and a few other minerals at the Chodarchay deposit in northwestern Iran.

The European Union and the global industry are very interested in the recovery processes of rare metals from major deposits as by-products. The micro-scale analysis with high-resolution techniques and the development of metallurgical processes will be fundamental. So, the study of the textural and chemical features of these minerals is the base for the development of extraction and recovery technologies.

Pyrite (FeS_2): The most common sulfide mineral within the crust is pyrite, existing extensively in magmatic-hydrothermal systems (Reich et al., 2013; Deditius et al., 2014; Dubosq et al., 2018). In addition, pyrite releases acid mine drainage producing potential damage to the environment (e.g., Sánchez-España et al., 2005; Santofimia et al., 2022). Pyrite is a significant Au host and occurs in various phases of Au mineralization (Fleet and Mumin, 1997; Reich et al., 2005; Zhang et al., 2014). The sulfur isotopic composition and in-situ trace element analysis of this mineral can imply the crystal growing past events and genesis of

pyrite and supply important data on the origin of mineralizing substances, mineralization fluids progression, and genesis of the deposit (Hu et al., 2019; Zhou et al., 2019; Chao et al., 2020). The trace element contents within pyrite can be representative of the constitution evaluating mineralization fluid and reconnaissance of the activation and deposit mechanism of elements throughout the fluid progression procedure (Agangi et al., 2014; Sykora et al., 2018; Zhou et al., 2019). Three types of trace elements occur inside pyrite: (1) structurally bound solid-solutions (Huston et al., 1995; Reich et al., 2013); (2) imperceptible nano fragments (Palenik et al., 2004; Deditius et al., 2011; Ciobanu et al., 2012); and (3) micro- to nano-size inclusions (Deditius et al., 2011; Sykora et al., 2018). The siderophile and chalcophile elements are dispersed in this sulfide. Arsenic, selenium, and antimony replace sulfur and can enter the pyrite lattice (Large et al., 2009; Cook et al., 2013; Agangi et al., 2014).

Chalcopyrite (CuFeS_2): In classic porphyry Cu deposits, chalcopyrite is mostly the principal host of numerous elements due to its plenty (Cook et al., 2011), but compared with other usual sulfides like sphalerite (Cook et al., 2009), chalcocite, bornite (Cook et al., 2011), and galena (George et al., 2015), it is usually regarded as a comparatively poor host for trace elements. The LA-ICP-MS element maps (George et al., 2016) displayed that when chalcopyrite co-crystallizes with sphalerite and galena, it is mostly the lowest favored host for a range of trace elements. Just Ag, As, Co, Se, Zn, and Pb were informed as solid-solutions within chalcopyrite, and Mn, In, Tl, Ga, and Hg are rare. Experimental gathering data for natural chalcopyrite in comparison with another sulfide mineral is rare, and in those cases where data were accounted, mostly the complete extent of elements was not examined, or numerous trace elements are too overhead to recognize precise concentration extents (Harris et al., 1984; Wohlgemuth-Ueberwasser et al., 2015; Sadati et al., 2016).

Sphalerite ($\text{Fe}_x\text{Zn}_{1-x}\text{S}$): Sphalerite is a significant host for an extensive range of minor and trace elements. The minor and trace element distribution within sphalerite is a significant basis for petrographic information, specifically for the determination of the genesis type and fluid origin of the ore body (Cook et al., 2009; Lin et al., 2011). It was suggested that reasonable contents of Ag can be inserted in the sphalerite lattice (e.g., Taylor and Radtke, 1969). Sphalerite is not a main host mineral for lattice-bound Au.

Galena (PbS): Galena has a simple formula and crystal structure, but some minor and trace elements can be included in its crystal lattice. Most of them (like Bi, Ag, Se, or Te) as by-products are extracted from galena-containing ore. Earlier investigations (Bethke and Barton, 1971; Tauson et al., 1986; Foord et al., 1988; Foord and Shawe, 1989; Liu and Chang, 1994; Lueth et al., 2000; Chutas et al., 2008; Renock and Becker, 2011) have recognized any minor and trace elements with various concentrations which can go into the galena crystal structure. The majority of published papers have focused on elements like Bi, Ag, or Sb (Van Hook, 1960; Foord et al., 1988; Foord and Shawe, 1989; Jeppsson, 1989; Lueth et al., 2000; Costagliola et al., 2004; Chutas et al., 2008; Renock and Becker, 2011), that exit partly at high concentration in some galena grains.

Chalcocite (Cu_2S): The chalcocite group minerals are remarkably distributed in various hydrothermally altered enrichment zones of sulfide Cu deposits. Macroscopic and microscopic methods are almost unidentifiable for distinguishing among members of the group because they have physical and chemical similarities. Chalcocite may include Ag, As, Au, Se, and especially Bi. The occurrence of these elements is connected to the crystal chemistry of the mineral (Cook et al., 2011).

Djurleite ($\text{Cu}_{31}\text{S}_{16}$): Djurleite, with a very complex structure, is a rare member of the chalcocite group. It is a secondary Cu sulfide mineral, and it is mainly detected with other Cu sulfides (e.g., chalcocite, bornite, chalcopyrite, or digenite) in the supergene enrichment zones of sulfide Cu deposits. Digenite, djurleite, and chalcocite are the Cu-rich members of the covellite to chalcocite series. Djurleite has a hexagonal close-packed structure with large unit cells (Evans, 1979).

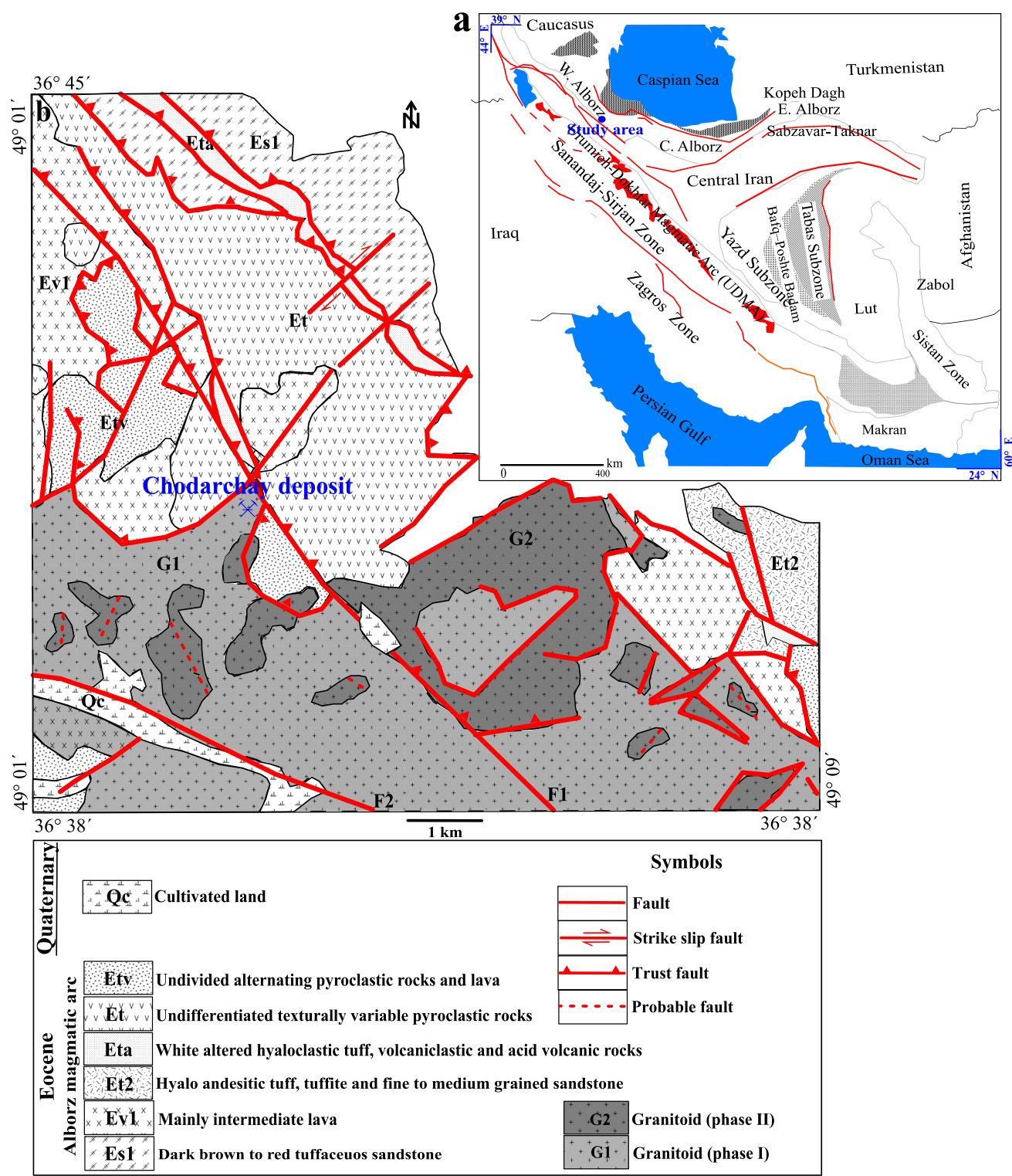


Fig. 1. (a) Position of the study area within the Alborz magmatic belt in the simplified structural map of Iran; (b) Distribution of the intrusive and volcano-volcanoclastic lithologic units in the Chodarchay area (based on Roudbar 1:100,000 geological map, after Nazari and Salamati, 1998). The location of the Chodarchay Cu deposit and faults are shown on the map (modified after Yasami et al., 2018).

Anilite (Cu_7S_4): Anilite was reported for the first time from Ani mine, Akita, Japan (Morimoto et al., 1969). Anilite is a product of alteration and oxidation changes series of chalcopyrite. These changes contain Cu/Fe and Cu/S ratio vacillation so that by Fe and S depletion and Cu enrichment, some new Cu minerals form. Alteration of chalcopyrite to covellite is accompanied by a Fe content decrease and simultaneous Cu increase. The following alteration of covellite to anilite

caused a Cu/S ratio increase from 0.9 in covellite to 1.77 in the anilite.

Bornite (Cu_5FeS_4): Following chalcopyrite, one of the most common Cu-(Fe)-sulfides is bornite. Bornite could include reasonable contents of Bi and Ag. Bornite is a comparatively poor Au host relative to other sulfides like chalcopyrite, pyrite, and pyrrhotite. Most exceptional Au contents in this sulfide can easily be traced to micron- and submicron inclusions (Cook et al., 2011).

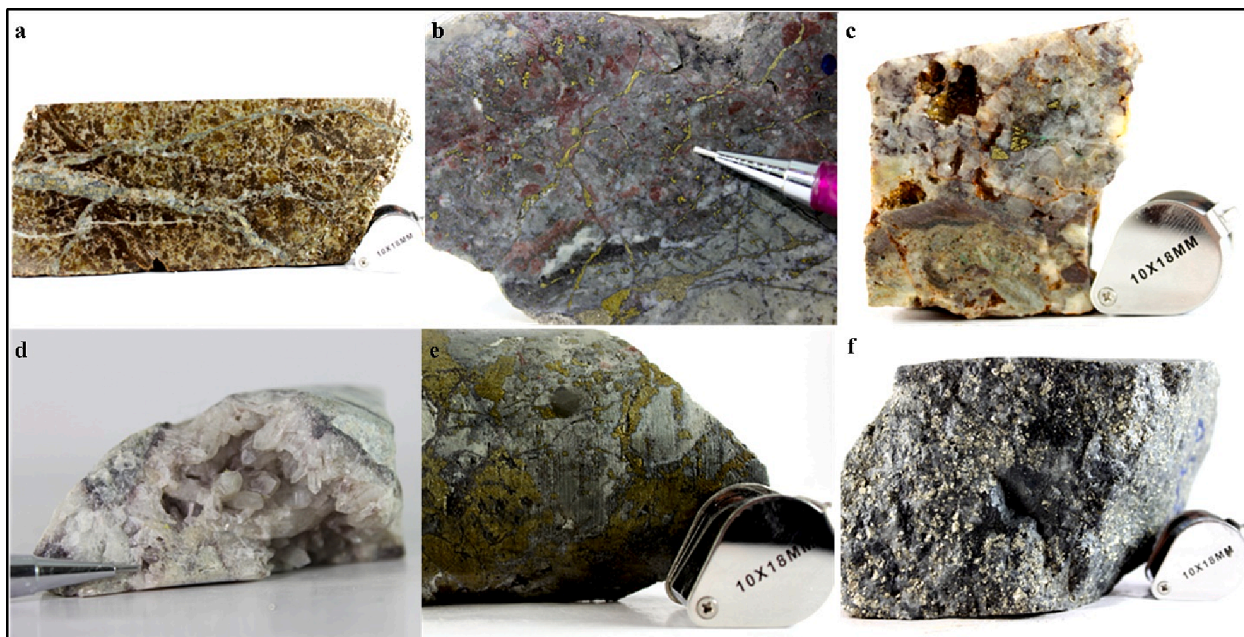


Fig. 2. The mineralization geometries of hand specimens in the Chodarchay deposit. (a) Silica vein-veinlets in alkali granite with sphalerite, galena, and pyrite mineralization (BH-CHD-0-1); (b) Chalcopyrite and pyrite vein-veinlets (BH-CHD-0-5); (c) Mineralized breccias with chalcopyrite and pyrite (BH-CHD-0-1); (d) Open space filling with euhedral quartz and minor chalcopyrite (BH-CHD-0-4); (e) Massive sulfides chalcopyrite (BH-CHD-0-5); (f) Dissemination texture of chalcopyrite and pyrite (BH-CHD-0-5).

Sulfosalts: Since there are a large number of minerals such as sulfosalts which are valuable sources for elements like Ag, Au, Bi, Cu, and Pb, the occurrence of sulfosalt minerals is important. The Bi sulfosalts provide crucial knowledge on the physical–chemical conditions of deposit formation and trace element distribution within ore deposits (Voudouris et al., 2013). Bismuth-bearing ore minerals can be signs of gold ore formation. The Au substitution into Bi melts and its crystallization as sulfosalt or telluride is a famous mechanism for Au aggregation and describes the paragenetic companionship of Bi minerals and Au in some ore deposits (Cook, 1997, 1998; Damian et al., 2000; Putz et al., 2003; Cook and Ciobanu, 2004; Ciobanu et al., 2006; Voudouris et al., 2013).

The principal purposes of this investigation are to determine sulfide and sulfosalt hosts for precious metals (like Ag and Au) and other valuable trace elements (strategic and critical) and to find the Ag partitioning trends among coexisting sulfosalts and between sulfosalts and sulfides at the Chodarchay ore deposit. Mineralogy, chemical composition, and associated assemblage of sulfosalts at Chodarchay are studied to provide information regarding the conditions of sulfosalt mineral formation and the genetic link between the abundance of sulfosalts. This study has evaluated associations between the contents of a certain element, whether one element's high concentration correlates with an increase or decrease in other elements, or whether the portioning of minor/trace elements in minerals relates to the geologic formation events of the deposit. This study, for the first time, describes the existence of various sulfosalts that were not informed before at Chodarchay.

2. Geological setting and characteristics of the Chodarchay deposit

The Tarom-Hashtjin subzone of the Alborz arc in NW Iran (Fig. 1a) hosts the Chodarchay deposit. This subzone was shaped in the Neo-Tethyan subduction back-arc. The Alborz was initiated in a Late Paleocene-Eocene extensional setting (e.g., Vincent et al., 2005; Verdel et al., 2011). The Alborz extensional structures have been inverted because of Arabia and Eurasia plates collision compression (early Oligocene) (e.g., Allen et al., 2003; Guest et al., 2007; Madanipour et al.,

2013). The spatial distribution of the deposits has been affected by this inversion. It could have caused overprinting of epithermal type on early porphyry mineralization. The Chodarchay NW-SE fault zone (F1) is accompanied by other W-E fault zone (F2) outside the deposit area (Fig. 1b). A normal status of the Chodarchay Fault motion preserves in the NW portion of the F1, implying two extensional and compressional mechanisms of the F1. The porphyry type is associated with the extensional mechanism, whereas the epithermal type is related to the compressional stage of the F1 Fault (Yasami et al., 2017).

Magmatic units contain intrusions (phases I and II) and volcano-volcaniclastic rocks. Phase I consists of quartz monzonite to quartz syenite, provided phase II principally includes granite and alkali granite. Petrological and geochemical studies of intrusions display high-K to shoshonitic affinity; they are I-type and metaluminous to moderately peraluminous. Both phases (I and II) emphasize subduction-related arc magmatism. The Chodarchay intrusions formed within the magmatic arc to post-collision settings (Yasami et al., 2018).

The mineralization is mainly hosted in the volcanic and volcanoclastic units (Eocene); some mineralization occurred within the quartz monzonite (Late Eocene). The porphyry type is mostly accompanied by a potassic alteration. The potassic alteration is continued by phyllic, argillic, and propylitic alteration types. Near the surface, the advanced argillic postdates other alteration types. The mineralization zones are separated into three zones from depth to surface:

- (1) Deeper hypogene sulfide zone: The principal hypogene sulfides include chalcopyrite, pyrite, sphalerite, and galena;
- (2) Near-surface enriched supergene sulfide zone: Chalcocite, djurite, anilite, digenite, and covellite exist in the supergene sulfide zone;
- (3) Surface leached and oxidized zone: After the exposition, the hypogene sulfides have been oxidized and replaced with Fe-(Mn) hydroxides, Cu-oxides, and Cu-carbonates like hematite, goethite, limonite, malachite, and azurite and form the oxide zone and leached cap (Yasami and Ghaderi, 2019).

The mineralization geometries include vein-veinlets, open space

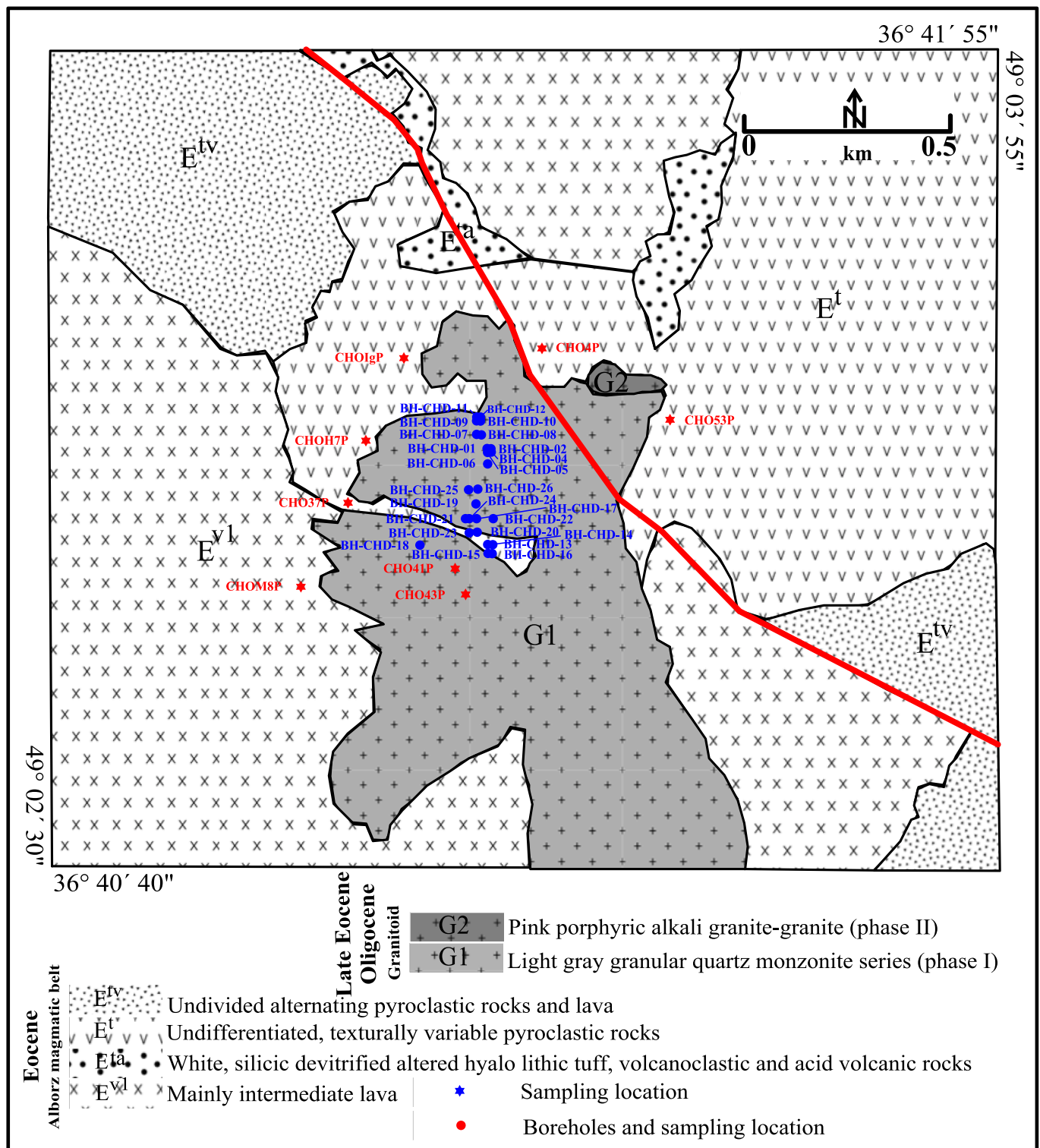


Fig. 3. The boreholes and sample locations plotted on the Chodarchay geology map (modified after Yasami et al., 2018b).

filling, mineralized breccias, dissemination, and massive sulfides (Fig. 2). The most volumetric mineralization geometry is the vein-veinlet with a few cm to several dm widths. The mineralized breccias are shown in outcrops and every of the logging cores. The disseminated mineral is principally pyrite. Pyrite, chalcocite, galena, sphalerite, chalcocite, bornite, anilite, djurleite, digenite, covellite, Cu, as well as Fe-(Mn) oxides, azurite, malachite, and sulfosalts are the major minerals.

Important primary control on mineralization at the Chodarchay is structural (Yasami et al., 2017), and little lithological control (Yasami and Ghaderi, 2019). Fluid boiling, cooling, and a mixture of magmatic with meteoric waters led to ore minerals settling (Mousavi Motlagh et al., 2021).

Table 1

Summary of the EPMA minor and trace element data (wt.%) for the hypogene and supergene sulfide ore minerals of the Chodarchay deposit.

	Ag	As	Au	Bi	Cu	Fe	Hg	Mn	Ni	Pb	S	Sb	Se	Te	Ti	Zn
Hypogene sulfide mineral																
Pyrite (N = 37)																
Min.	0	0	0	0.11	0	41.65	0	0	0	0.04	51.86	0	0	0	0	0
Max.	0.05	1	0	0.8	1.1	46.88	0.16	0.05	0.47	0.94	53.92	0	0.07	0.05	0.03	0.36
Avg.	0.006	0.1	0	0.23	0.11	45.63	0.03	0.005	0.16	0.25	52.55	0	0.006	0.004	0.005	0.06
Chalcopyrite (N = 58)																
Min.	0	0	0	0	33.21	28.32	0	0	0	0	32.39	0	0	0	0	0
Max.	0.4	0.11	0	0.88	35.78	30.98	0.17	0.79	0.5	2.14	34.96	0.25	0.13	0.08	0.03	1.51
Avg.	0.04	0.01	0	0.19	34.34	29.60	0.04	0.02	0.11	0.31	34.06	0.01	0.02	0.007	0.004	0.17
Sphalerite (N = 24)																
Min.	0	0	0	0	0	0.27	0	0	0	0	28.61	0	0	0	0	62.78
Max.	0.1	0.09	0.05	0.29	1.43	4.27	0.19	0.23	0.39	0.27	33.89	0.04	0.06	0.03	0.03	69.78
Avg.	0.017	0.014	0.004	0.14	0.267	1.244	0.017	0.082	0.03	0.12	32.21	0.003	0.007	0.004	0.005	65.49
Galena (N = 23)																
Min.	0	0	0	0.25	0	0.02	0	0	0	82.22	10.74	0	0	0	0	0
Max.	1.15	0.07	0	2.89	1.39	0.3	0.07	0.08	0.66	86.93	13.2	0.06	0.92	0.4	0.04	0.33
Avg.	0.26	0.01	0	1.01	0.14	0.18	0.003	0.01	0.08	84.91	12.64	0.005	0.07	0.04	0.01	0.05
Supergene sulfide mineral																
Chalcocite (N = 6)																
Min.	0.04	0	0	0.09	78.4	0.13	0	0	0	0.17	20.25	0	0	0	0	0
Max.	0.09	0	0	0.23	79.48	0.53	0.07	0.01	0.06	0.8	20.89	0.02	0.07	0.01	0.01	0.14
Avg.	0.07	0	0	0.15	78.75	0.28	0.025	0.003	0.01	0.50	20.54	0.005	0.04	0.002	0.002	0.033
Djurleite (N = 2)																
Min.	0.3	0	0	0.32	76.42	0.1	0	0	0.09	0.46	20.6	0.01	0	0	0	0.07
Max.	0.33	0	0	0.49	77.05	0.29	0.11	0	0.11	0.58	20.92	0.06	0.06	0.07	0	0.11
Avg.	0.31	0	0	0.40	76.73	0.19	0.55	0	0.1	0.52	20.76	0.035	0.03	0.035	0	0.09
Anilite (N = 1)																
0	0	0	0	0.03	74.79	1.85	0	0	0	0.37	20.58	0	0.05	0.02	0	0
Bornite (N = 2)																
Min.	0	0	0	0.09	63.06	4.81	0	0	0	0.14	23.45	0	0	0	0	0
Max.	0.23	0	0	0.21	69.04	15.79	0.05	0	0.08	0.31	29.42	0	0.03	0	0.02	0
Avg.	0.11	0	0	0.15	66.05	10.3	0.025	0	0.04	0.225	26.43	0	0.015	0	0.01	0

3. Materials and methods

3.1. Sampling and microscopic study

Following the collection of samples from the field and logging cores and preparing microscopic sections (N = 57) of representative ore samples from the Chodarchay deposit, a total of 20 suitable thin-polish sections were studied. Firstly, the sulfides (e.g., pyrite, chalcopyrite, galena, and sphalerite), sulfosalts, and some other ore minerals were studied using optical reflected light microscopy (OM), and the ore minerals paragenesis and 247 selected points within the samples for electron microprobe (EPMA) investigation were determined. A Nikon microscope with a magnification of up to 100X at Tarbiat Modares University, Tehran, Iran, was used for the study and to take reflected light images of the selected microscopic sections.

3.2. Electron microprobe analysis

Electron microprobe analyses were performed at the Iranian Mineral Processing Research Center (IMPRC) facility in Karaj, Iran. The elemental analyses were done using a Cameca X100 electron microprobe operated at an acceleration potential of 20 keV, a probe current of 20nA, and a beam diameter of 1–5 μm . Therefore, reflected light microscopy was used together with EPMA to recognize and characterize the general texture of the samples, mineral relationships, and possible overprinting features, as well as for the determination of contents of the trace and minor elements within the Chodarchay ore minerals. EPMA can also demonstrate that sometimes high concentrations of trace elements in a mineral are afforded by micro- or nano-inclusions of another mineral instead of the trace elements in the mineral lattice.

3.3. Fire-assay analysis

Thirty samples were taken from different sulfide and oxide parts of

the logging cores of the Chodarchay deposit to check the presence of Au in various sections. The collected samples were sent to the Zarazma Laboratory, Tehran, Iran, for Au by fire-assay analyses. Fire-assay is a total-recovery method. The method includes the fusion of the samples at an 1100 °C temperature, throughout which gold is segregated from gangue and gathered within a molten Pb. Then Pb is eliminated by cupellation, afresh at high temperature, to lead to a prill of Au. The prill is solved in acids. The Zarazma's proprietary assay flux mix is planned to provide the best recuperation from the extended range of rock types feasible. The highest excellence assay consumables (flux, crucibles, and cupels) originated from surrounding the world to certain that assay integrity is guaranteed. The Zarazma's standard assay charge is 25 g.

3.4. ICP-OES analysis

Thirty-eight core and surface samples were processed by Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) analyses for a series of elements that contained Ag, As, Ba, Cd, Cu, Hg, Mn, Mo, Ni, Pd, S, Sb, W, and Zn at the Iranian Mineral Processing Research Center (IMPRC) facility in Karaj, Iran. The ICP results were employed for the analysis of economic ore formation and also for describing geochemical properties in the system.

The boreholes and sample locations on the Chodarchay geology map have been shown in Fig. 3.

4. Results

4.1. Mineralization and EPMA analytical results

During this study, EPMA was taken to analyze the contents of the interesting minor and trace elements in sulfide, sulfosalts, and some other ore minerals. Much information was collected from these elemental concentrations. A summary of concentration data for each interesting element for each sulfide mineral is displayed in Table 1.

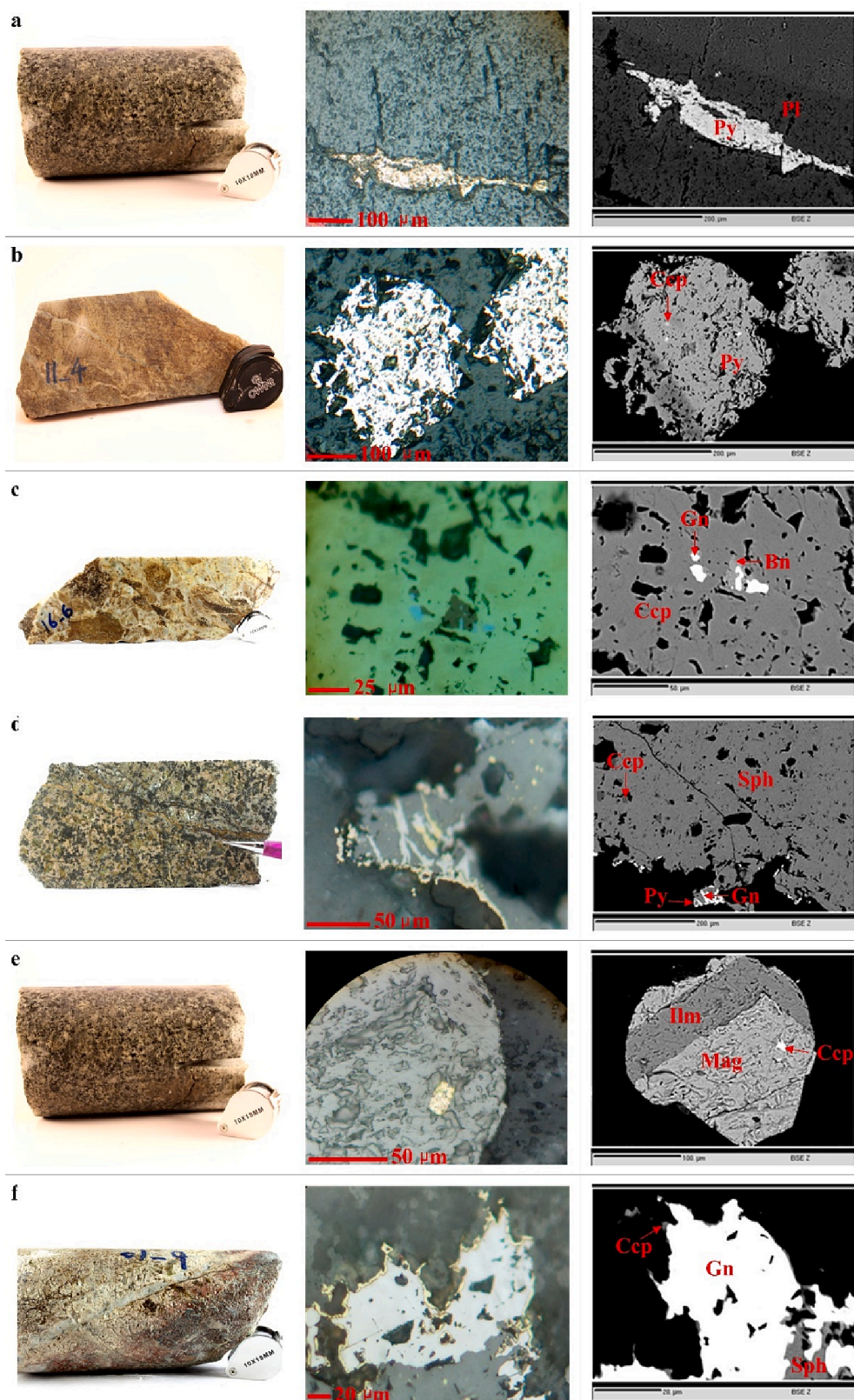


Fig. 4. Hand specimen, thin and polished-section photomicrographs under the petrographic microscope, and EPMA photo micrograph. (a) Alteration of pyrite in quartz monzonite that replaced plagioclase (CHO24-6P); (b) Pyrite associated with chalcopyrite inclusion in rhyolitic crystal tuff (CHO11-3P); (c) Bornite and galena inclusions inside chalcopyrite in hydrothermal breccias (CHO16-5P); (d) Chalcopyrite disease in sphalerite associated with replacing galena and marginal pyrite in quartz monzonite (CHO15-18P); (e) Chalcopyrite inclusion in disseminated magnetite in quartz monzonite (CHO24-6P); (f) Galena replaced the sphalerite and chalcopyrite replaced galena in rhyolitic crystal tuff (CHO1-8P). Abbreviations; Bn: Bornite; Ccp: Chalcopyrite; Gn: Galena; Ilm: Ilmenite; Mag: Magnetite; Pl: Plagioclase; Py: Pyrite; Sph: Sphalerite.

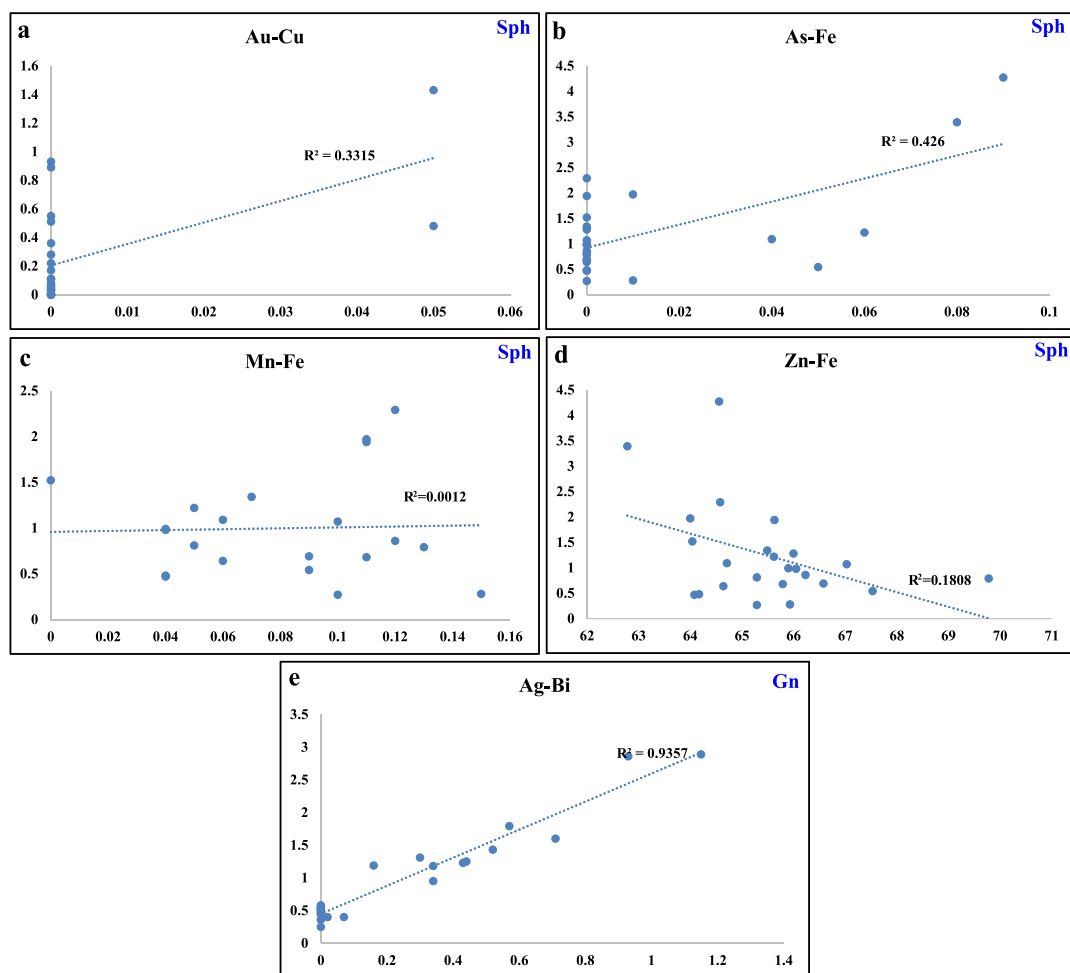


Fig. 5. Positive correlation of (a) Au with Cu; (b) As with Fe; (c) No correlation of Mn with Fe; Negative correlation of Zn with Fe; (d) in the sphalerite (e) Ag with Bi in the galena based on EPMA analyses.

4.1.1. Sulfide ore minerals

4.1.1.1. Hypogene sulfides. Pyrite: Pyrite is one of the sulfides in the Chodarchay ore body and is usual in most of the altered and mineralized rocks. There are at least two pyrite generations (Fig. 4a-b). Alteration-related pyrite crystals are euhedral to subhedral disseminated crystals. Mineralization stage pyrites are euhedral to subhedral crystals together with other sulfide minerals, including chalcopyrite, sphalerite, and galena minerals.

Pyrite was analyzed in 37 points, and the results are shown in Table 1. Gold and antimony have not been recorded in the pyrite minerals. Silver concentrations are very low; those are above the detection limit values only in 7 spots of analysis (maximum 0.05 wt%). Pyrite is the host for Cu, As, Pb, Bi and Zn. Their highest amounts can reach up to 1.1 wt%, 1 wt%, 0.94 wt%, 0.8 wt%, and 0.36 wt%, respectively. Arsenic and Cu have changeable concentrations within the studied pyrite samples. Among the analyzed elements, As, Bi, and Pb are the most plentiful within this sulfide mineral (Table 1).

Chalcopyrite: Chalcopyrite is the most plentiful sulfide ore mineral at Chodarchay. There are at least two generations of chalcopyrite (Fig. 4b-f). The first generation is together with other ore minerals (usually pyrite, galena, and sphalerite). The second generation is formed by inclusions within the sphalerite (chalcopyrite disease; Fig. 4d). Chalcopyrite was analyzed in 58 points (Table 1). The chalcopyrite at Chodarchay contains S at 32.39–34.96 wt%, Fe at 28.32–30.98 wt%, and Cu at 33.21–35.78 wt%. Chalcopyrite minerals also have trace amounts of Pb, up to 2.14 wt%; Bi, up to 0.88 wt%; Mn, up to 0.79 wt%;

Zn, at 0–1.51 wt% and Ag, up to 0.4 wt%. Other trace elements such as As, Hg, Sb, and Se are present in comparatively low concentrations. Similar to the pyrite grains, no elemental association was identified in the chalcopyrite crystals.

Sphalerite: Sphalerite occurred in the mineralized breccias and quartz veinlets alone or associated with other sulfides (Fig. 4d, f). Sphalerite was analyzed in 24 points (Table 1). Iron is the most usual minor element within sphalerite (marmatite) and exists in all sphalerites analyzed in this study. Microprobe analyses display Fe contents in the 0.27–4.27 wt% range, with an average amount of 1.24 wt%. Manganese in the component of sphalerite reaches up to 0.23 wt%. Copper was concentrated in sphalerite (up to 1.43 wt%). The abundance of Ag is not constant (from 0 to 0.1 wt%), occurring in nine spots. Among sulfide minerals, only sphalerite contains Au. Most of the spots across all analyzed samples described Pb and Bi up to 0.27 and 0.29 wt%, respectively. Mercury was found to be above detection limits in some samples. Antimony was found in four spots with low concentrations. Selenium is absent from sphalerite (only four spots have been above the detection limit). Comparison of the abundance in wt.% of some interesting elements compared with each other shows that a mildly positive correlation of Au with Cu and a moderately strong positive correlation exists between As and Bi (Fig. 5a, b). No correlation exists between Mn and Fe (Fig. 5c). The correlation of Fe with Zn is -0.42 (Fig. 5d).

Galena: Like sphalerite, galena occurs in the quartz veinlets and mineralized breccias alone or together with other sulfides (Fig. 4c, d, f). Galena was analyzed in 23 points (Table 1). Bismuth has the maximum quantified concentrations within galena compared with any minor and/

Table 2

Summary of some minor and trace element EPMA data (wt.%) for sulfosalt minerals of the Chodarchay deposit.

	Ag	As	Au	Bi	Cu	Fe	Hg	Mn	Ni	Pb	S	Sb	Se	Te	Ti	Zn
Hodrushite (N = 1)																
	1.77	1.21	0	21.44	28.83	4.94	0	0.02	0	6.45	18.62	6.71	0.26	0	0.04	3.96
Watanabeite (N = 1)																
	0.08	6.96	0	2.3	43.35	1.09	0	0	0.07	0.65	26.63	6.68	0.18	0	0.01	8.96
Colusite (N = 2)																
Min.	0.1	6.48	0	0.52	42.42	7.1	0.02	0	0	0.7	28.5	6.13	0.11	0	0	6.36
Max.	0.1	5.39	0	0.42	43.2	7.04	0.04	0	0	0.31	28.8	5.95	0.14	0	0	6.73
Avg.	0.1	5.93	0	0.47	42.81	7.07	0.03	0	0	0.50	28.7	6.04	0.12	0	0	6.545
Friedrichite (N = 2)																
Min.	0.39	0	0	22.64	19.23	0.63	0	0	0.01	18.18	19.89	0	0.1	0	0	0
Max.	0.73	2.11	0	26.52	25.74	14.93	0.01	0.03	0.19	18.75	24.6	2.36	0.19	0.05	0.02	3.07
Avg.	0.56	1.05	0	24.58	19.23	14.93	0.005	0.015	0.1	18.46	22.24	1.18	0.14	0.02	0.01	1.535
Aikinite (N = 1)																
	0.73	0	0	20.96	21.42	14.65	0.06	0	0	17.51	13.82	0	0.09	0	0.02	0
Paderaitite (N = 2)																
Min.	0.1	0	0	18.1	34.9	0.92	0	0	0	5.06	21.23	0.06	0.04	0	0	0
Max.	0.38	4.25	0	29.07	35	9.82	0	0.02	0.19	9.18	21.98	4.29	0.14	0	0	5.85
Avg.	0.24	2.12	0	23.58	34.95	5.37	0	0.01	0.1	7.12	21.60	2.17	0.09	0	0	2.92
Emplectite (N = 15)																
Min.	0.06	0	0	15.89	31.32	0.01	0	0	—	0	16.55	0	0	0	0	0
Max.	7.53	2.83	0.06	41.31	42.61	2.73	0.14	0.07	—	6.28	21.17	15.53	1.48	0.41	0.04	7.06
Avg.	1.45	0.39	0.007	34.62	37.85	0.61	0.03	0.012	—	0.8	18.93	2.48	0.25	0.04	0.01	1.56
Cuprobismuthite (N = 2)																
Min.	3.52	0.1	0	32.39	22.99	0.56	0	0	0	12.11	17.67	1.65	0.39	0	0	1.39
Max.	3.99	0.53	0	34.19	23.85	0.84	0.01	0	0	15.12	18.44	4.78	0.43	0.76	0.01	2.54
Avg.	3.75	0.31	0	33.29	23.42	0.7	0.005	0	0	13.61	18.05	3.21	0.41	0.38	0.005	1.96
Neyite (N = 10)																
Min.	0	0	0	29.95	8.4	0.05	0	0	0	29.46	14.52	0	0.01	0	0	0
Max.	2.74	0.06	0.08	39.81	12.27	4.83	0.08	0.07	0.23	36.55	16.8	0.01	0.36	0.3	0.04	0.17
Avg.	0.48	0.006	0.025	34.59	10.45	1.79	0.02	0.014	0.04	34.35	15.89	0.001	0.214	0.03	0.013	0.06
Cupropavonite (N = 1)																
	9.71	0	0	45.85	15.04	1.11	0	0.02	0	10.45	16.29	0	0.49	0	0	0.1
Makovickyite (N = 2)																
Min.	0	0	0	53.95	7.26	0.54	0	0	0	12.15	17.43	0.01	0.35	0	0.03	0
Max.	0.1	0	0	57.36	10.43	2.64	0	0.01	0.21	15.86	17.83	0.06	0.37	0.04	0.05	0
Avg.	0.05	0	0	55.65	8.845	1.59	0	0.005	0.1	14.00	17.63	0.035	0.36	0.02	0.04	0
Galenobismutite (N = 1)																
	0.02	0	0	49.85	5.95	0.22	0.04	0	0	20.95	17.88	0	0.36	0	0.02	0.02
Gustavite (N = 2)																
Min.	7	0	0	41.26	7.77	0	0	0	0	18.73	16.35	0	0.3	0	0	0
Max.	7.74	0.04	0.11	47.47	7.86	0.2	0	0	0	24.55	16.36	0	0.42	0.2	0.03	0.07
Avg.	7.37	0.02	0.055	44.36	7.81	0.1	0	0	0	21.64	16.35	0	0.36	0.1	0.015	0.035
Heyrovskyite (N = 1)																
	4.33	0	0	26.93	5.01	0.38	0	0.02	0	47.87	12.43	0.09	0.41	0.35	0	0.03

or trace element analyzed. The Bi contents change by twelve orders of magnitude among the samples. Silver has the highest amount among sulfide minerals in the galena. Galena with elevated Ag and Bi is reported (up to 1.15 and 2.89 wt%, respectively). The Cu concentrations within galena change from the low detection limit to 1.39 wt%. About two-thirds of the spots show Cu concentrations below the detection limit. Selenium, Te, Zn, and Fe were detected with a maximum of 0.92, 0.4, 0.33, and 0.3 wt%, respectively. The Sb concentration in most of the samples is zero; exceptions are three spots in which record Sb values are observed (0.01, 0.04, and 0.06 wt%). Gold could not be detected in the galena. Results show an absolutely strong positive correlation of Ag with Bi within galena (Fig. 5e).

4.1.1.2. Supergene sulfides. Chalcocite: Chalcocite has been replaced with chalcopyrite in fractures and margins. Chalcocite is characterized by the supergene sulfide zone. The chalcocite was analyzed in 6 points. The chalcocite contains mostly Cu at 78.4–79.48 wt% and S at 20.25–20.89 wt% as well, chalcocite concedes measured amounts of up to 0.8 wt% Pb, Fe 0.53 wt%, 0.23 wt% Bi, 0.14 wt% Zn, with minor 0.04–0.09 wt% Ag (Table 1).

Djurleite: Djurleite is associated with galena, pyrite, and chalcopyrite. It replaced the chalcopyrite in the margins. The djurleite was detected in two points and contained Cu at 76.42–77.05 wt% and S at 20.6–20.92 wt%. Djurleite is enriched in Pb (0.58 wt%), Bi (0.49 wt%),

Ag (0.33 wt%), Fe (0.29 wt%) and Zn (0.11 wt%). The average Ag content in this mineral is 0.31 wt%. Thus, it is an Ag-bearing djurleite (Table 1).

Anilite: This secondary Cu-sulfide mineral is associated with native Cu and supergene secondary Cu-Fe-bearing minerals. Anilite was determined at one point with 74.79 wt% Cu and 20.58 wt% sulfur and yielded measured values of 1.85 and 0.37 wt%, respectively for Fe and Pb (Table 1).

Bornite: Bornite is associated with chalcopyrite and partially replaces it (Fig. 4c). It was analyzed in 2 points. The bornite contains Cu at 63.06–69.04 wt%, S at 23.45–29.42 wt% and Fe at 4.81–15.79 wt%. Lead, Ag, and Bi display measured amounts up to 0.31, 0.23, and 0.21 wt %, respectively (Table 1).

4.1.2. Sulfosalts

A summary of the concentration data for some of the interesting elements for each determined sulfosalt mineral from the Chodarchay deposit is shown in Table 2.

Hodrushite (Cu₈Bi₁₂S₂₂): Hodrushite occurred inside chalcopyrite associated with cuprobismuthite (Fig. 6a). This sulfosalt was found in one point with 28.83 wt% Cu, 21.44 wt% Bi, 18.62 wt% S, 6.71 wt% Sb, 6.45 wt% Pb. Measured Ag is 1.77 wt% (Table 2).

Paderaitite (AgPb₂Cu₆Bi₁₁S₂₂): Paderaitite has been formed in the margin of chalcopyrite associated with colusite and cupropavonite that

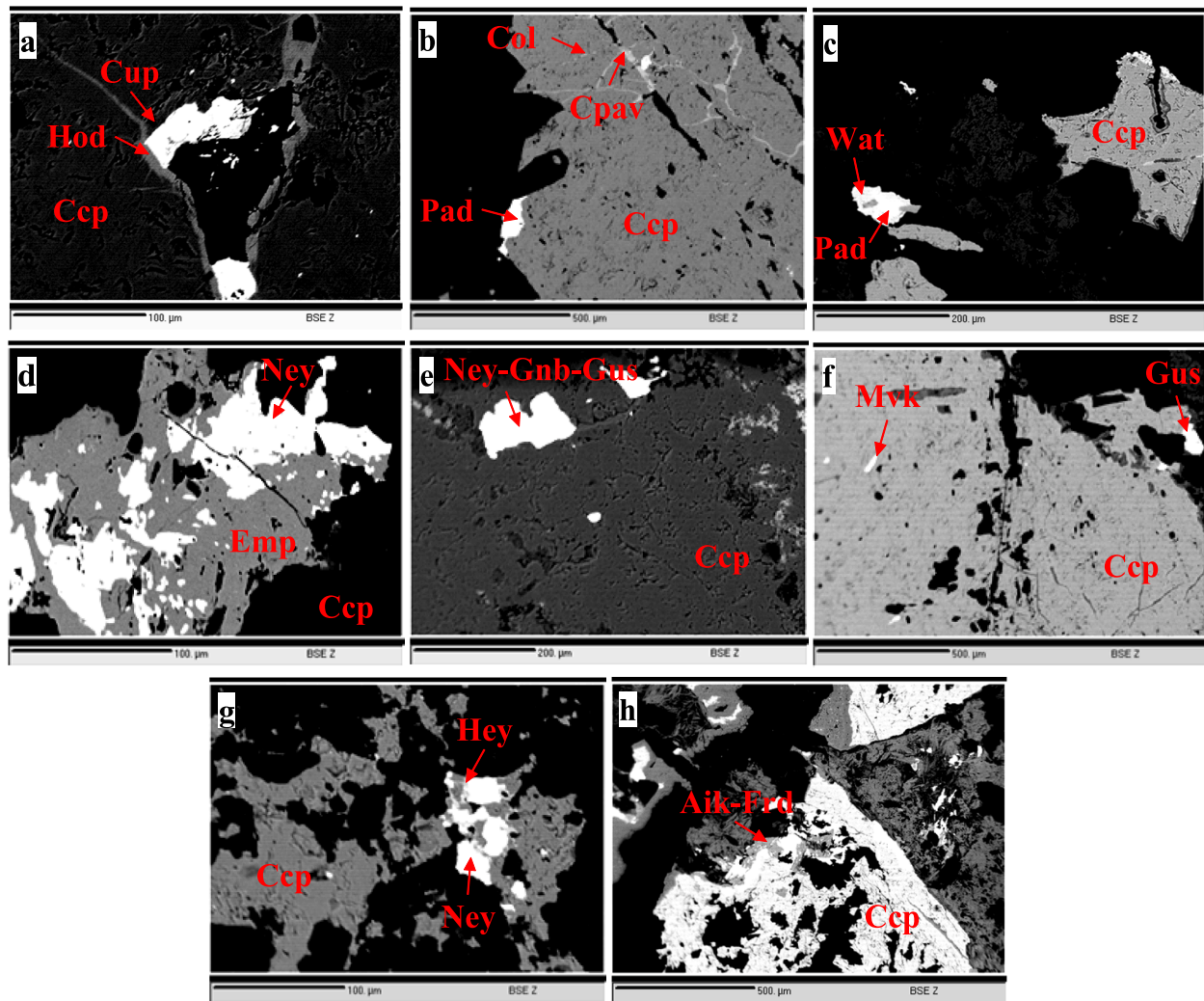


Fig. 6. EPMA photomicrographs. (a) Hodrushite and cuprobismuthite inside chalcopyrite (CHO20-4P); (b) Paderaitite in the margin of chalcopyrite; and colusite and cupropavonite inside chalcopyrite (CHO20-4P); (c) Inclusions of paderaitite inside watanabeite (CHO20-4P); (d) Association of neyite and emplectite in chalcopyrite in quartz monzonite (CHO15-18P); (e) Neyite, galenobismutite, and gustavite (as series) inclusion in chalcopyrite in breccias (CHO20-4P); (f) Marginal gustavite and makovickyite inclusions in chalcopyrite in breccias (CHO26-8P); (g) Neyite and heyrovskyite in its margin in chalcopyrite in breccias (CHO5-3P). (h) Aikinite-friedrichite in margin of chalcopyrite (CHO26-8P). Abbreviations; Aik: Aikinite; Ccp: Chalcopyrite; Col: Colusite; Cpav: Cupropavonite; Cup: Cuprobismuthite; Emp: Emplectite; Gnb: Galenobismutite; Gus: Gustavite; Hey: Heyrovskyite; Hod: Hodrushite; Mvk: Makovickyite; Ney: Neyite; Pad: Paderaitite; Wat: Watanabeite.

replaced chalcopyrite (Fig. 6b) and occurred inside and associated with watanabeite too (Fig. 6c).

It was detected in two points with a maximum of 35, 29.07, 21.98, 9.16, and 9.82 wt% for Cu, Bi, S, Pb, and Fe, respectively. Silver content reached 0.38 wt% (Table 2).

Emplectite (CuBiS_2): Emplectite occurred associated with neyite inside chalcopyrite (Fig. 6d). Emplectite was analyzed in 15 points. This mineral contains both Au and Ag. Of course, Au has been recorded in only two points with very low values (0.05 and 0.06 wt%). The amount of Ag in this mineral is significant, reaching 7.53 wt% (average 1.45 wt%). Following cupropavonite and gustavite, it has the third position in having Ag among sulfosalts. The average Bi in this mineral is 34.62 wt%.

Cuprobismuthite ($\text{Cu}_{10}\text{Bi}_{12}\text{S}_{23}$): Cuprobismuthite occurred inside chalcopyrite associated with hodrushite (Fig. 6a). It was analyzed in two points with Bi, Cu, S, and Pb contents up to 34.19, 23.85, 18.41, and 15.12 wt%, respectively. The maximum amount of Ag is about 4 wt% (Table 2).

Neyite ($\text{AgCu}_3\text{Pb}_{12.5}\text{Bi}_{13}\text{S}_{34}$): Neyite occurred as the inclusion or as intergrown with emplectite (Fig. 6d), associated with gustavite and galenobismutite (as series; Fig. 6e) and heyrovskyite (Fig. 6g) inside chalcopyrite. Neyite was analyzed in 10 points. It contains Bi at

29.95–39.81, Pb at 29.46–36.55, S at 14.52–16.8, and Cu at 8.4–12.27 wt%. The maximum of Ag is 2.75 wt%. This sulfosalt shows concentrations of Au in three points (at 0.06 to 0.08 wt%; Table 2).

Cupropavonite ($\text{AgPbCu}_2\text{Bi}_5\text{S}_{10}$): Cupropavonite associated with colusite replaced chalcopyrite (Fig. 6b). This mineral was analyzed at one point. The content of Bi, S, Cu, and Pb is 45.81, 16.29, 15.01, and 10.45 wt%, respectively. Cupropavonite has the highest amount of Ag (9.71 wt%; Table 2).

Makovickyite ($\text{Ag}_{1.5}\text{Bi}_{5.5}\text{S}_9$): The makovickyite occurs as inclusion in the chalcopyrite (Fig. 6f).

It was detected in two points. The makovickyite in Chodarchay contains Bi at 53.95–57.36, S at 17.43–17.83, Pb at 12.15–15.86, and Cu at 7.26–10.43 wt%. Ag content is up to 0.1 wt% (Table 2).

Galenobismutite (PbBi_2S_4): The neyite-galenobismutite-gustavite (as series) inclusion occurred in the chalcopyrite (Fig. 6e). This was determined at one point with 49.85 wt% Bi, 20.95 wt% Pb, 17.88 wt% S and about 6 wt% Cu. Ag content is low (0.02 wt%; Table 2).

Gustavite ($\text{AgPbBi}_3\text{S}_6$): There is gustavite as the inclusion in chalcopyrite (Fig. 6e) and also occurred in the margin of chalcopyrite (Fig. 6f). The gustavite was analyzed in two points with Bi at 41.26, Pb at 18.73–24.55, S at 16.35–16.36 and Cu at 7.77–7.86 wt%. The gustavite

Table 3

Fire-assay analyses of gold contents from the surface to the depth along with the core numbers and typical minerals in hand specimens on 30 samples from the Chodarchay deposit. Abbreviations: Ccp: Chalcopyrite; Py: Pyrite; Sph: Sphalerite; Gn: Galena; Bn: Bornite; Jrs: Jarosite.

	Sample no.	Core no.	Approximate depth (m)	Hand specimen description	Typical ore minerals	Au (ppb)
1	CHO12-0P	BH-CHD-0-12	2.70	Breccia	Sph-Gn-Ccp-Py-Oligiste	125
2	CHO19-0P	BH-CHD-0-19	5.20	Veinlet in alkali granite	Ccp, Fe-Mn oxides	101
3	CHOA18-0P	BH-CHD-0-18	6.00	Quartz monzonite	Fe oxides	5
4	CHO19-5P	BH-CHD-0-19	7.20	Veinlet in alkali granite	Jrs, Fe-Mn oxides	207
5	CHOA1-0P	BH-CHD-0-1	9.00	Hydrothermal breccia	Fe-Mn oxides	121
6	CHOA5-0P	BH-CHD-0-5	9.20	Breccia tuff	Fe-Mn oxides	253
7	CHO1-2P	BH-CHD-0-1	9.40	Vuggy quartz	Fe oxides	90
8	CHO5-1P	BH-CHD-5	10.50	Veinlet	Sph-Gn-Ccp-Py-Bn-sulfosalt	97
9	CHOA17-0P	BH-CHD-0-17	11.10	Alkali granite	Fe oxides	5
10	CHO26-8P	BH-CHD-0-26	12.60	Breccia	Ccp-Py-sulfosalt, Fe oxides	170
11	CHOA6-1P	BH-CHD-0-6	14.50	Breccia	Fe oxides	80
12	CHO2-3P	BH-CHD-0-2	17.10	Breccia	Sph-Gn	98
13	CHOA17-1P	BH-CHD-0-17	18.70	Veinlet	Oligiste	227
14	CHO17-3P	BH-CHD-0-17	19.5	Veinlet	Oligiste	87
15	CHO17-7P	BH-CHD-0-17	20.70	Veinlet	Ccp, Oligiste	138
16	CHO5-2P	BH-CHD-0-5	21.00	Veinlet	Py-Ccp-Sph	434
17	CHOA4-1P	BH-CHD-0-4	21.70	Tuff	Oligiste	15
18	CHOA17-2P	BH-CHD-0-17	22.70	Breccia	Fe oxides	47
19	CHO17-9P	BH-CHD-0-17	23.70	Veinlet	Fe oxides	32
20	CHO15-17P	BH-CHD-0-15	33.40	Quartz monzonite	Gn-Sph-Ccp-Py	18
21	CHO5-3P	BH-CHD-0-5	35.00	Breccia	Ccp-Py-sulfosalt	227
22	CHO13-4P	BH-CHD-0-13	36.10	Quartz monzonite	Oligiste, Sph	11
23	CHOA10-2P	BH-CHD-0-10	39.60	Andesite	Ccp, Oligiste	5
24	CHOA9-1P	BH-CHD-0-9	54.00	Tuff	Sph-Gn-Oligiste	37
25	CHO1-11P	BH-CHD-0-1	56.50	Alkali granite	Sph-Gn-Ccp	92
26	CHOA12-2P	BH-CHD-0-12	56.50	Alkali granite	Gn	10
27	CHO15-10P	BH-CHD-0-15	63.60	Breccia	Ccp, Oligiste	5
28	CHO1-14P	BH-CHD-0-1	66.30	Alkali granite	Ccp-Bn-sulfosalt	65
29	CHOA6-0P	BH-CHD-0-6	75.00	Breccia	Fe oxides	20
30	CHOA5-2P	BH-CHD-0-5	97.40	Alkali granite	Gn	5

with 7.74 wt% Ag and 0.11 wt% Au has a second and first rank, respectively, among all sulfide and sulfosalt ore minerals (Table 2).

Heyrovskyite ($\text{Pb}_{10}\text{AgBi}_5\text{S}_{18}$): The heyrovskyite was surrounded by neyite inclusion in chalcopyrite (Fig. 6g). The heyrovskyite contains 47.87 wt% Pb, 26.93 wt% Bi, 12.43 wt% S, 5.01 wt.%Cu and the Ag content 4.33 wt% (Table 2).

Watanabeite ($\text{Cu}_4(\text{AsSb})_2\text{S}_5$): This mineral has been detected at one point associated with paderaitite inclusions (Fig. 6c). The Au content in

this ore mineral is zero. The amount of Ag is negligible. The amounts of arsenic and antimony are equal to 6.96 and 6.68, respectively. This mineral contains 43.35 wt% Cu. The amount of Zn reaches 8.96 wt% (Table 2).

Colusite ($\text{Cu}_{12}\text{V}(\text{SbAsSn})_3\text{S}_{16}$): The colusite and cupropavonite occurred inside of chalcopyrite (Fig. 6b). This mineral does not include Au and its average Ag concentration is 0.1 wt%. The Cu concentration extends up to 43.2 wt%. The average concentrations of As and Sb are

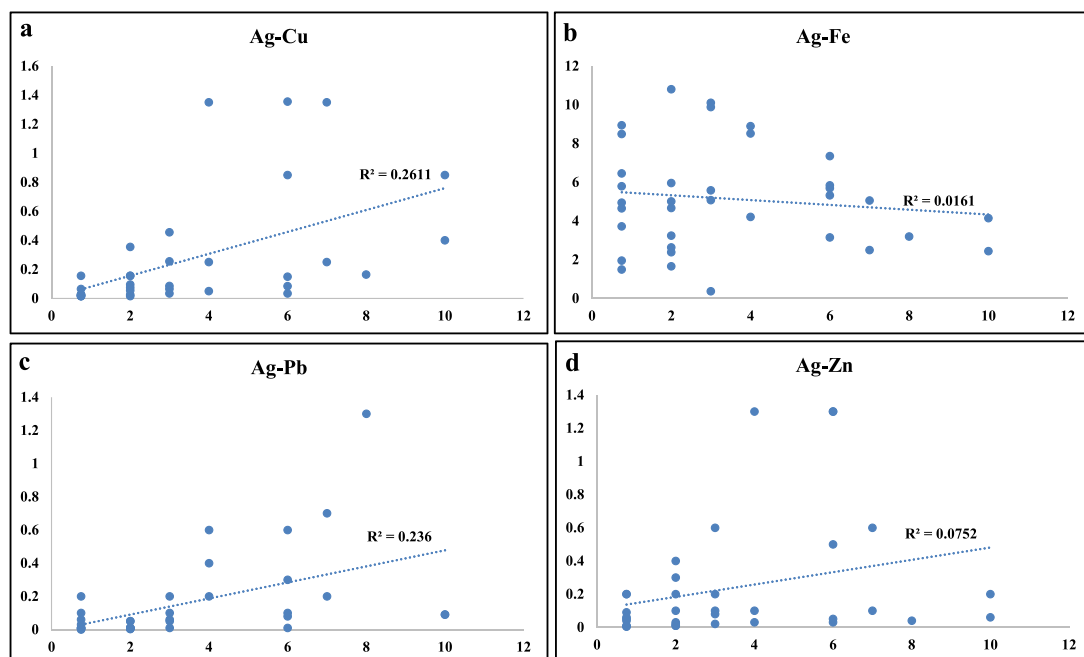


Fig. 7. The correlation coefficients between Ag with (a) Cu, (b) Fe, (c) Pb, and (d) Zn based on ICP-OES analyses. The highest correlation of Ag is with Cu.

Table 4

ICP-OES analyses of some elements (Cu, Fe, Pb, and Zn) contents along with Ag from various outcrop and drill core samples of the Chodarchay deposit (N = 38). Lithology, ore mineral paragenesis, and sampling depth are noted for each sample. Abbreviations: Ccp: Chalcopyrite; Py: Pyrite; Sph: Sphalerite; Gn: Galena; Bn: Bornite; Mlc: Malachite; Cu: Native Cu; Pm: Psilomelane; Pyr: Pyrolusite; Gth: Goethite.

	Sample no.	Core no.	Lithology	Depth	Ore mineral paragenesis	Ag (µg/g)	Cu (%)	Fe (%)	Pb (%)	Zn (%)
1	CHO01-0P	BH-CHD-00-1	Rhyolitic tuff	0.2	Ccp, Py, Cc, Mag, Hem, Mal, Gth	4	1.35	4.21	0.4	0.1
2	CHO1-1P	BH-CHD-0-1	Hydrothermal breccia	7.5	Py, Ccp, Mlc	8	0.17	3.19	>1	0.04
3	CHO1-8P	BH-CHD-0-1	Rhyolitic crystal tuff	31.7	Ccp, Py, Gn, Sph	2	0.07	5.01	0.05	0.3
4	CHO1-11P	BH-CHD-0-1	Alkali granite	59.3	Gn, Sph, Ccp, Py, sulfosalt	4	0.3	8.52	0.6	>1
5	CHO2-1P	BH-CHD-0-2	Tuff	13.6	Ccp, Py, Sph, Gn	7	1.35	5.05	0.2	0.1
6	CHO3-0P	BH-CHD-0-3	Rhyolitic crystal tuff	41.6	Ccp, Py, Gn, Sph	3	0.04	0.37	0.1	0.02
7	CHO4-4P	BH-CHD-0-4	Alkali granite	36.5	Ccp, Py, Gn, Sph, Bn	2	0.02	10.8	0.01	0.4
8	CHO5-20P	BH-CHD-0-5	Trachyte	97	Ccp, Py, Gn	<1	0.01	6.45	0.01	0.06
9	CHO6-0P	BH-CHD-0-6	Breccia	8	Ccp,Mlc	10	0.9	2.44	0.09	0.06
10	CHO10-1P	BH-CHD-0-10		16.5	Cu, Cc	<1	1.35	8.49	0.1	0.2
11	CHO10-4P	BH-CHD-0-10	Rhyolite	32.8	Sph, Gn, Ccp, Py	6	0.9	3.15	0.6	>1
12	CHO12-2P	BH-CHD-0-12	Breccia	27.8	Ccp, Py, Gn, Sph, Sulfosalt	<1	0.1	5.79	0.04	0.2
13	CHO12-6P	BH-CHD-0-12	Alkali granite	56.8	Py, Sph	6	0.09	7.35	0.06	>1
14	CHO13-2P	BH-CHD-0-13	Tectonic breccia	34	Ccp, Py, Sph	2	0.09	5.96	0.01	0.1
15	CHO14-6P	BH-CHD-0-14	Breccia	67.3	Ccp, Py, Sph, Gn	10	0.51	4.14	0.08	0.2
16	CHO15-17P	BH-CHD-0-15	Alkali granite	66.5	Ccp, Py, Gn, Sph	3	0.08	10.1	0.2	0.2
17	CHO16-5P	BH-CHD-0-16	Hydrothermal breccia	28.2	Ccp, Py, Sulfosalt, Gn, Bn	3	0.2	5.07	0.07	0.08
18	CHO17-1P	BH-CHD-0-17	Alkali granite	14	Hem, Py, gth	6	0.04	5.32	0.3	0.5
19	CHO17-4P	BH-CHD-0-17	Silica veinlet	21.3	Ccp, Py, Mlc	6	1.35	5.69	0.1	0.05
20	CHO17-7P	BH-CHD-0-17	Breccia	25.2	Ccp, Py, Cc, Bn	27	1.35	6.5	0.02	0.06
21	CHO18-1P	BH-CHD-0-18	Quartz syenite	6.2	Hem, Gth, Pm, Ccp, Py	3	0.06	5.58	0.01	0.6
22	CHO18-4P	BH-CHD-0-18	Quartz monzonite	17	Cc	7	0.3	2.49	0.7	0.6
23	CHO18-5P	BH-CHD-0-18	Quartz monzonite	26	Mag, Ilm, Ccp, Py	<1	0.02	4.65	0.08	0.04
24	CHO19-0P	BH-CHD-0-19	Silica veinlet	2.8	Ccp, Py, Mlc	4	0.1	8.89	0.2	0.03
25	CHO19-1P	BH-CHD-0-19	Alkali granite	4.5	Hem, Gth, Mag, Ilm, Pm, pyr, Py, Ccp	3	0.4	9.88	0.05	0.1
26	CHO19-10P	BH-CHD-0-19	Quartz monzonite	34	Sph, Gn, Ccp	<1	0.02	1.49	0.2	0.09
27	CHO20-4P	BH-CHD-0-20	Alkali granite	51.2	Ccp, Py, Sph, Gn, Sulfosalt	70	1.35	16.1	0.1	0.1
28	CHO22-0P	BH-CHD-0-22	Alkali granite	24.2	Mag, Hem, Ccp, Sph	2	0.05	2.63	0.01	0.03
29	CHO24-6P	BH-CHD-0-24	Quartz monzonite	59.1	Mag, Ilm, Ccp, Py	2	0.1	4.67	0.005	0.01
30	CHO4P	—	Aplite	0	Mag	<1	0.02	8.94	<0.001	0.01
31	CHO5P	—	Intrusion	0	Mag, Ilm, Ccp	<1	0.01	1.95	0.01	0.005
32	CHO37P	—	Breccia	0	Ccp	25	1.35	5.79	0.2	0.4
33	CHO41P	—	Quartz monzonite	0	Mag, Ilm, Ccp, Py	<1	0.06	4.94	0.01	0.05
34	CHO43P	—	Quartz monzonite	0	Hem, Py, Ccp	2	0.3	3.24	0.01	0.02
35	CHO53P	—	Silicified unite	0	Pm	2	0.1	2.38	0.06	0.2
36	CHOH7P	—	Breccia	0	—	6	0.2	5.84	0.01	0.03
37	CHOIgP	—	Rhyolite	0	—	2	0.01	1.66	0.003	0.01
38	CHOM8P	—	Lithic tuff	0	Mag	<1	0.01	3.72	0.004	0.004

5.93 and 6.04 wt%, respectively.

Aikinite–bismuthinite series (aikinite (PbCuBiS₃) and friedrichite (Pb₅Cu₅Bi₇S₁₈)): Aikinite and friedrichite were found a margin of chalcopyrite and are rare (Fig. 6h). Aikinite was detected in one and friedrichite tow points. They do not have Au but contain up to 0.73 wt% Ag.

4.2. Fire-assay analysis

Based on fire-assay analyses, the highest and lowest contents of Au are 434 and 5 ppb, respectively. The highest value is related to the BH-CHD-5 core and taken from a depth of 21 m with a pyrite-chalcopyrite-sphalerite paragenesis sample (Table 3).

4.3. ICP-OES analysis

ICP-OES analysis was carried out on 38 surface and core samples for 28 elements. The deepest sample was taken from the trachyte unit at a depth of 97 m. The Ag concentration in this sample was below the detection limit (1 μg/g) of the device. The highest content of Ag measured is equal to 70 μg/g, was taken from a depth of about 51 m and the alkali granite, followed by two values of 27 and 25 μg/g, both from the breccia unit showing the highest Ag contents in the samples, respectively from a depth of about 25 m and the surface. In all three samples with the highest Ag contents, the Cu content is more than 1 %. Despite the low Bi content in most of the samples that could not be measured, the Bi content in the three mentioned samples is high, and in

the sample that contains 70 μg/g of Ag, Bi reaches more than 1 % (1566 μg/g). The correlation coefficients between Ag with Cu, Fe, Pb, and Zn are + 0.51, −0.13, +0.49, and + 0.27, respectively. The highest correlation of Ag is with Cu (Fig. 7). In Table 4, some element contents along with Ag, and the characteristics of the samples, including their lithology and depth, are given.

5. Discussion

One of the principal aims of this study is to try to recognize any elemental associations; elemental comparisons were made for all of the main minor elements.

5.1. Trace element distribution in the sulfides, mineralization fluids sources, and processes

The trace element content of sulfides in porphyry systems depends on the element concentrations in the ore-carrying fluids (Zhu et al., 2016), elemental partitioning conduct into the mineral/fluid, and the existence of co-crystallizing sulfides. Thus, the physicochemical condition changes in the mineralizing fluids and the thermodynamic mineral assemblages' stability probably cause trace element variations of the sulfides. Examination of trace element contents and micro-textural features of sulfides/ sulfosalts helps to clarify the mineralizing processes.

Pyrite: Pyrite is present in most hydrothermal systems, and its precipitation efficiently rules the partitioning of economically and

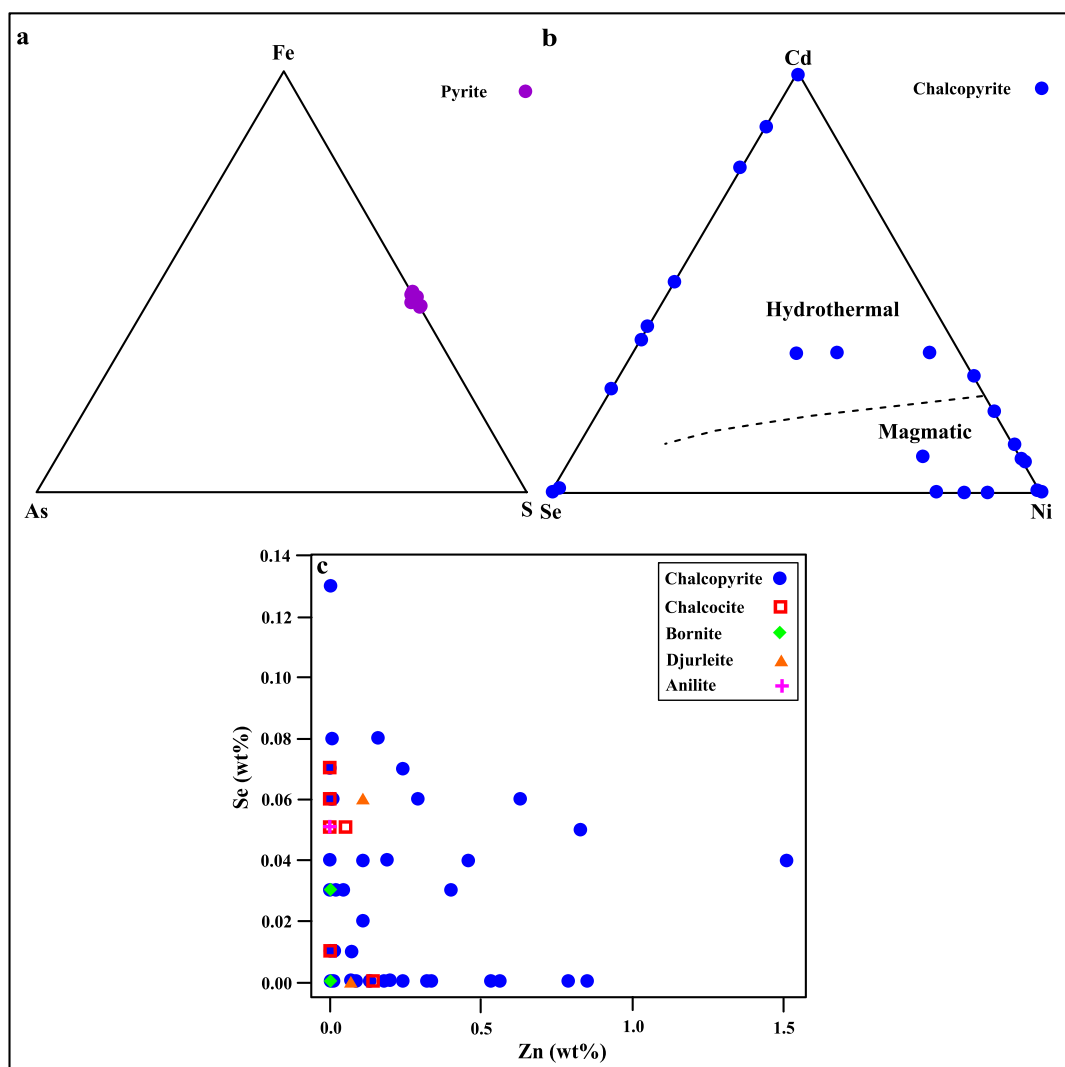


Fig. 8. (a) The As-Fe-S ternary diagram for pyrite from the Chodarchay ore deposit; (b) Chodarchay chalcopyrite on the Se-Ni-Cd ternary diagram proposed by Duran et al. (2019) for discrimination between chalcopyrite from hydrothermal and magmatic sulfide deposits; (c) Plot showing Zn and Se variation in coexisting Cu-Fe sulfides at Chodarchay.

environmentally important trace elements like Ag, As, Au, and heavy metals (Deditius et al., 2008; Large et al., 2009). Pyrite usually includes several notable metallurgical trace elements. Three Cu, Pb, and Ni elements are bounded within the pyrite lattice. As and Se replace S and enter the pyrite lattice (Large et al., 2009; Cook et al., 2013; Agangi et al., 2014). Despite the low average concentrations of arsenic in the pyrite (Fig. 8a), it is the second host for arsenic at the Chodarchay deposit after sulfosalts. Low As contents in the pyrite cause relatively clean pyrites with few minor elements. Extremely few observations have recorded high Cu contents within pyrite (recognized as cuprian pyrite). It has been agreed that most Cu within pyrite places as Cu-sulfides (such as chalcopyrite) inclusions; the scrutinizers have stated that Cu in considerable amounts can be placed in the pyrite lattice so that in some cases achieving wt.% levels (Frenzel and Ottemann, 1967; Einaudi, 1968; Clark, 1970; Radcliffe and McSween, 1970; Pacevski et al., 2008). The excited concentrations of Cu within Chodarchay pyrites, which extend up to 1.1 wt%, imply that pyrite can function as a significant Cu scavenger in Cu deposits and so act as an unexpected control on Cu throughout ore system development. Lead, because of the big ionic size, is least probably to be entering into the pyrite lattice; Pb sulfide has a more rapid precipitation rate than Fe in fluid (Huerta-Diaz and Morse, 1992; Koglin et al., 2010). Hence, galena precipitates previously, so it is usually present as an inclusion within pyrite (Zhao et al., 2011). Pyrite

fractures can act as appropriate traps so that remobilized Pb can combine with available S and form nanoparticles of galena (Owen et al., 2018). As soon as the recrystallization of pyrite, this Pb-phase becomes impressively contained within the pyrite or has linear Pb-bearing characteristics within the pyrite crystals (Rollog et al., 2019; Rollog et al., 2020). The existence of Pb within Chodarchay pyrite composition (up to 0.94 wt%) may be pointed out to the impress of Pb remobilization and Pb phase's inclusions formation at a sub-microscopic level during pyrite recrystallization. Pyrite can act as a physical trap for remobilized Ag, Bi, and Te. Maximum contents of Ag and Te are very low (both up to 0.05 wt%) in the Chodarchay pyrite. However, Bi reaches 0.8 wt%. Nickel as an isomorphism form can readily partition into the pyrite crystal structure and does not easily exist even in the re-crystallization process of pyrite (Huerta-Diaz and Morse, 1992; Luther, 1998; Tribouillard et al., 2006; Large et al., 2009; Koglin et al., 2010). Therefore, the Ni content of pyrite may supply knowledge about the mineralizing fluid source. Usually, the mantle and its dependent mafic-ultramafic units have enriched Ni, but felsic units have low Ni contents (Palme and O'Neill, 2003; Zhao et al., 2011). The Ni contents within the Chodarchay pyrite (up to 0.47 wt%) are more than continental crust Ni (19–60 µg/g; Rudnick and Gao, 2003), so it is sourced from mantle mafic-ultramafic rocks. Geochemical features suggested a chemically enriched lithospheric mantle origin for the Chodarchay intrusions (Yasami et al.,

2018).

Chalcopyrite: Chalcopyrite is one of the most common sulfides in ore deposits, and its trace element composition changes with the deposit type. These changes are mainly controlled by the geological setting and its crystal structure, making chalcopyrite a potential indicator sulfide mineral for exploration (Duran et al., 2018; George et al., 2018). Chalcopyrite is mostly thought of as a comparatively poor host for trace elements relative to other common sulfide minerals like sphalerite (Cook et al., 2009) and galena (George et al., 2015). Thus, its capacity for trace element hosting rises in the lack of other co-crystallizing sulfide minerals (George et al., 2016). Chalcopyrite can receive a range of trace elements in its tetragonal structure, like Ag, Bi, Cd, Co, Ga, Hg, In, Mn, Pb, Sb, Se, Sn, and Zn (Pauling and Brockway, 1932; Butler and Nesbitt, 1999; George et al., 2018). Substitution of minor/trace elements within the chalcopyrite lattice is complicated, and various mechanisms, such as coupled replacements, have been assessed (e.g., Belissont et al., 2019). These coupled substitutions involve monovalent (Cu^+ and Ag^+), divalent (Zn^{2+} , Cd^{2+} , and Pb^{2+}), trivalent (Fe^{3+} , In^{3+} , and Sb^{3+}), and/or tetravalent (Se^{4+} , Sn^{4+} , Bi^{4+} , and Ge^{4+}) ions. The Chodarchay chalcopyrite shows trace enrichments of Pb (to 2.1 wt%) and Zn (to 1.51 wt%) that can be a divalent coupled substitution. Bismuth (up to 0.88) probably displays tetravalent coupled substitution. Chalcopyrite has the second most Ag (up to 0.4 wt%) among sulfide minerals after galena, and it may be a monovalent coupled substitution.

The Se-Cd-Ni ternary diagram displays that chalcopyrite from both magmatic and hydrothermal types exist at Chodarchay (Fig. 8b). Systematic changes in Cd/Zn ratio forcefully appertain on crystallization temperature (George et al., 2018). Therefore, the Cd/Zn ratio in hydrothermal chalcopyrite is usually applied to detect changes in physicochemical conditions pending mineral crystallization. Non-constant Cd/Zn ratio in the Chodarchay chalcopyrite (range between 0 and 6) may indicate varying physicochemical conditions throughout its crystallization. A high Cd/Zn ratio (0.13) is generally attributed to higher crystallization temperatures (George et al., 2018). Other than the values equal to zero, most Chodarchay chalcopyrite samples have high Cd/Zn ratios and show higher temperatures of crystallization.

Sphalerite: Iron is the major cation that substitutes Zn sites in the sphalerite lattice and reaches up to 4.27 wt% in the Chodarchay sphalerite. No correlation exists between Mn and Fe (Fig. 5c). This indicates that Mn and Fe compete with each other in substituting for Zn within sphalerite. The correlation of Fe with Zn is -0.42 (Fig. 5d). Negative associations of Fe (divalent cations) with Zn within sphalerite signify an isovalent substitution for Zn^{2+} (Cook et al., 2009; Keith et al., 2014).

Copper does not easily fractionate into the sphalerite structure and often occurs within sphalerite as fine chalcopyrite inclusions, well known as 'chalcopyrite disease' (Barton and Bethke, 1987; Keith et al., 2014). Therefore, the variations in Cu concentrations are mainly because of changes in the chalcopyrite contamination density rather than real lattice-bound Cu, particularly where there are large variations in Cu contents. In the Chodarchay sphalerite crystals, Cu concentrations vary from 0 to 1.43 wt% (Table 1). This data is confirmed with microscopic evidence for chalcopyrite inclusions within the sphalerite (see Fig. 4d).

The sphalerite samples show Ag occurrence, and sphalerite can be an Ag host at this deposit. There is no correlation between Ag and other elements that suggests Ag may be a solid-solution element in the sphalerite. Sphalerite is the only sulfide mineral containing Au at Chodarchay. Sphalerite has 0 to 0.05 wt% Au and has a positive correlation with Cu (about $+0.6$) (Fig. 5a). Thus, Au did not exist as a solid-solution within sphalerite but maybe as a nano-separate mineral phase.

Minor and trace element concentrations of sphalerite in various deposit types are different because the fractionation of these elements into sphalerite is affected by crystallization temperature, metal origin, and the sphalerite content of the ore body. Sphalerite supplies an applicable tool to account for precipitation temperatures according to its Fe

content, in addition to its Fe/Zn ratio (Keith et al., 2014). The Fe/Zn ratio in the Chodarchay sphalerite minerals is from 0.004 to 0.07 with ~ 0.02 wt% average. The temperature-sensitive ratio of Fe/Zn in sphalerite shows an evolution from high-temperature (up to 280°C) to low-temperature ($<220^\circ\text{C}$) precipitation and sulfidation state decreasing.

Galena: An excellent association between Ag and galena is detected in many hydrothermal ore deposits (Costagliola et al., 2004). Our data display that galena samples are poor in Sb (up to 0.06 wt% with an average of ~ 0.005 wt%). The Sb/Bi ratio within galena is up to 0.15 wt%, which is an indicator of relatively high-temperature deposits (Malakhov, 1968). The substitution of Pb with Ag (Sb, Bi) elements is prevented at lower than 300°C (Chutas et al., 2008).

The measured Bi and Ag can be in solid-solutions and the consequence of μm inclusions of Bi-Ag bearing phases. The high contents of Ag (up to 1.5 wt%) in the Chodarchay galena (the highest among sulfide minerals) can be described with the concurrent Bi enrichment (up to 2.89) because Ag can enter into galena as an outcome of coupled substitution with Ag + Bi for 2 Pb atoms. There is a very high correlation ($R = 97$) between Ag and Bi (Fig. 5c) through the Chodarchay data. The coupled substitution $\text{Ag}^+ + (\text{Bi})^{3+} \leftrightarrow 2\text{Pb}^{2+}$ is approved by the achieved data. Silver and Bi have the highest calculated contents within galena of any analyzed element; thus, Ag and Bi can be economically extracted as by-products from the Chodarchay galena.

Cadmium was entered into galena; the simple isovalent substitution ($\text{Cd}^{2+} \leftrightarrow 2\text{Pb}^{2+}$) can be concluded. The analytical results for the sample from Chodarchay display that Cu (to 1.39 wt%) is most probably in solid-solution within the galena lattice. Both the selenium (up to 0.92 wt%) and tellurium (to 0.4 wt%) are situated within the crystal lattice of galena in this study.

Coexisting galena and sphalerite: There are coexisting galena and sphalerite in some of the Chodarchay samples in this study. So allows the collation of minor and trace element partitioning between these coexisting sulfide minerals. Silver and Bi are mainly partitioned within the Chodarchay galena, while Hg and Cd are concentrated in the Chodarchay sphalerite. Most of the analyzed points in the Chodarchay sphalerite samples have relatively low Pb, supporting Cook et al. (2009) suggestion that because of ionic size differences between Pb and Zn, sphalerite is an unsuitable host for Pb and where galena is present, Pb will most likely enter into this mineral. Copper contents are higher in the sphalerite than in the galena, revealing that sphalerite is the preferred Cu host relative to the galena at Chodarchay. Selenium is absent from sphalerite (only four spots have been above the detection limit). In contrast, the galena shows Se mostly above the detection limit. So this can be proposed that in galena-sphalerite assemblage, Se is preferentially entered into galena.

Supergene copper sulfides: The chalcocite and djurleite in the current study contain a concentration of Ag, maximum to 0.09 and 0.23 wt%, respectively. Silver is absent in anilite. The highest Bi content among these three minerals belongs to djurleite (up to 0.49 wt%), but the highest concentrations of Pb (0.8 wt%) and Zn (0.14 wt%) among these three minerals belong to chalcocite. Thus, djurleite at Chodarchay can be Ag-Bi host and chalcocite for Pb. The bornite in this investigation is enriched in Bi (0.21 wt%), Pb (0.31 wt%), and Ag (0.23 wt%). Bornite can be one of the major carriers of Ag and Bi.

The Cu-(Fe) sulfides are common in both hypogene and supergene Cu ore deposits and are useful Cu ore minerals. To know where hypogene processes influence Cu-(Fe)-sulfides and their surface characteristics have usages for ore mineral processing and environmental geochemistry (Vaughan et al., 1987; Harmer et al., 2005). Numerous agents can influence the portioning of trace elements within Cu-(Fe) sulfides, involving physicochemical factors of the ore-bearing fluids, trace element partitioning among coexisting sulfides, substitution in the mineral structure in contrast to mineral inclusions, and the Cu-(Fe) sulfide paragenesis (Ciobanu et al., 2017; Martin et al., 2019). Therefore, a comparison of the trace element distribution can display features that rein the trace element distribution in various Cu-(Fe) sulfides.

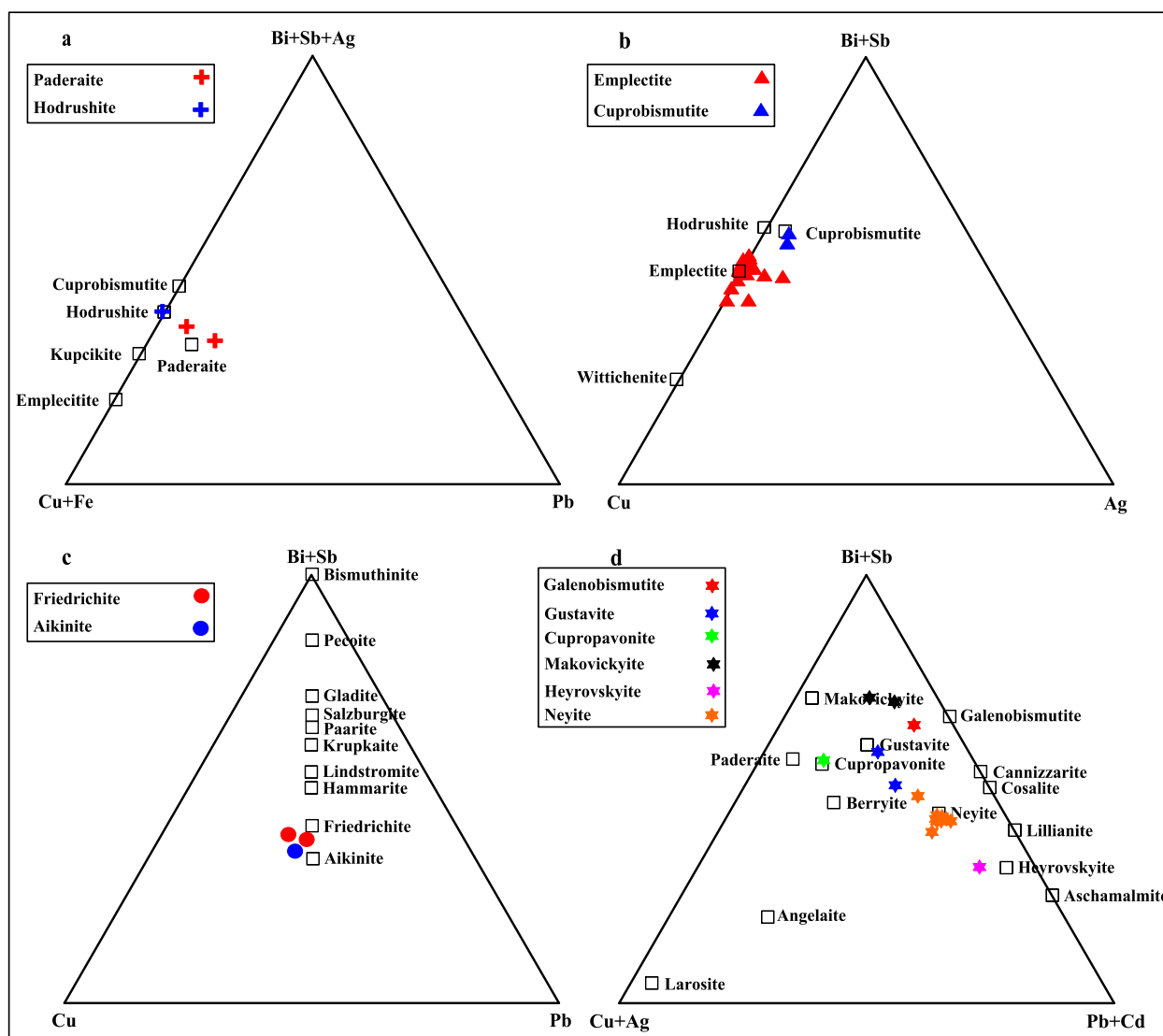


Fig. 9. Ternary diagrams of Bi-sulfosalts for the Chodarchay samples: (a) Cuprobismuthite group minerals; (b) Cu-Bi-S minerals; (c) Aikinite–bismuthinite series; (d) Pb-Bi-Sb minerals.

In general, trace element distribution is importantly distinct among Cu-(Fe) sulfides. The trace element contents are also disparate in the Chodarchay Cu-(Fe) sulfides (chalcopyrite, chalcocite, bornite, djurleite, and anilite). However, the contents of Zn and Se within every Cu-(Fe) sulfide exhibit a tiny change (Fig. 8c), revealing that none of the Zn and Se is rather entered into chalcopyrite, bornite, anilite, djurleite, or chalcocite. Other trace elements like Ag and Bi imply considerable variations within the Cu-(Fe) sulfides. The Ag content is low in chalcocite and some chalcopyrite points but is high in the djurleite and bornite, which indicates that this element is preferentially entered into chalcopyrite, djurleite, and bornite. Silver is a low-temperature element, so this can illustrate the increasing trend of the Ag concentration within the Cu-(Fe) sulfides in the investigation. The existence of precious metals in these sulfides is a crucial concept for most mining operations, which can economically raise the worth of the Cu deposit. Thus, Ag can be extracted as a valuable by-product. Also, there are some damaging elements (e.g., Zn, Pb, and Bi) in Cu-(Fe) sulfides that reduce the total grade of the Cu concentration.

5.2. Trace element distribution in the sulfosalt minerals

The resulting data from Chodarchay confirm the presence of a

complex mineralogical paragenesis. All sulfosalt minerals at Chodarchay are observed in chalcopyrite. They show the typical exsolution texture of the sulfosalts from chalcopyrite (see Fig. 6). The record of sulfosalts within the examined chalcopyrite is consistent with the extended range of trace elements like Ag, As, Bi, and Sb (Table 1).

Bismuth–sulfosalts (since Bi is the dominant cation, they are called Bi-sulfosalts) discovered in this study include cuprobismuthite group minerals (cuprobismuthite, hodrushite, and paderaitite), aikinite–bismuthinite series (aikinite and friedrichite), emplectite, makovickyite, galenobismuthite, gustavite, heyrovskyite, neyite, and cupropavonite; and As-Sb bearing Cu sulfosalts (watanabeite and colusite). This number of sulfosalt minerals ($N = 14$) is rare in a mine or deposit in Iran's epithermal deposits, and this is the first time that it has been reported from a deposit in the Tarom-Hashtjin subzone and Alborz zone. There are no detailed modern analyses for most epithermal deposits in the Tarom region. The Bi–sulfosalts compositions can be plotted in the ternary spaces (Fig. 9). Most of the high Bi contents were reported from spots with anomalously high Pb and/or Cu contents. This implies that all of the high Bi data resulted from Cu-Pb-Bi sulfosalts. The late-stage Bi-rich assemblages involve emplectite, gustavite, galenobismuthite, makovickyite, and heyrovskyite. The emplectite occurrence characterizes the Te and Bi enrichments in the hydrothermal fluids.

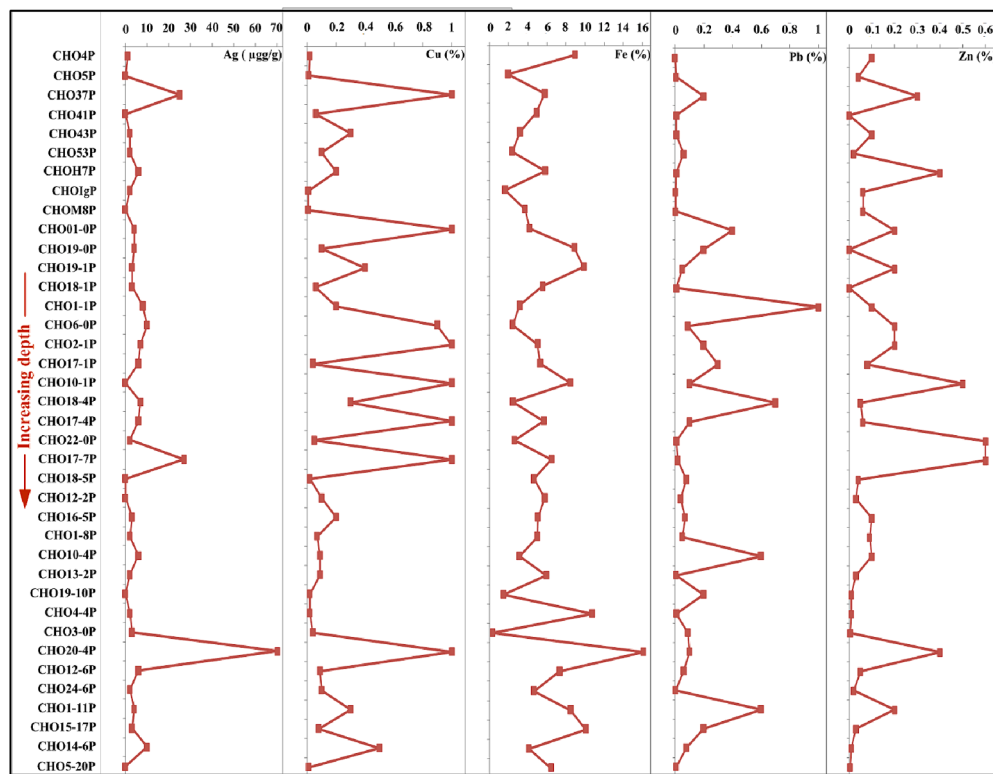


Fig. 10. Distribution of Ag ($\mu\text{g/g}$), Cu (%), Fe (%), Pb (%), and Zn (%) from surface to depth in the ICP-OES analytical results of 38 samples from outcrop and different drill cores of the Chodarchay deposit. The depth at which the samples are collected is noted in Table 4.

Heyrovskyite contains Ag (+Cu) concentration via the exchange of $\text{Ag} + \text{Bi} \leftrightarrow 2\text{Pb}$ (Makovicky et al., 1991; Cook, 1997).

Some sulfosalt species host lattice-bound Au, and some cannot accept Au. This transition works as a crystal lattice limit for Au. The Au–Ag occurrence was distinguished within the neyite, emplectite, and gustavite (Table 2). Silver did not only seem contained in the sulfide minerals (galena contains up to 1.15 wt% Ag) but also was concentrated in sulfosalts (up to 9.71 wt% Ag in the cupropavonite), a mineral that contains up to 45.85 wt% Bi. The Ag-rich ore minerals crystallize from fractionated ore-bearing fluids that have successively been cooled during migration from the source toward the surface, which is mostly related to the emplacement of the intrusions. Textural relationships between chalcopyrite and Ag-carrier sulfosalts suggest that they could be exsolution via sub-solidus re-equilibration from pre-existing Ag-bearing sulfides in the cooling system. The Bi sulfosalt group minerals mainly consist of Cu, Pb, Bi, and S, with minor components Ag, maintaining the charge balance.

Bismuth sulfosalts are important indicators of physicochemical conditions due to being sensitive relative to temperature, Eh–pH, and oxygen and sulfur fugacities (Cook and Ciobanu, 2004). Cook (1997) related $> 350^\circ\text{C}$ for Bi assemblages. Consideration of the available experimental data on the cuprobismuthite group points to crystallization temperatures of more than 300°C (Sugaki et al., 1981). At higher than 350°C , major values of Sb can enter into heyrovskyite (Chang et al., 1980). However, at Chodarchay, Sb concentration in heyrovskyite is negligible. Therefore, due to the presence of these minerals at Chodarchay, the mineralization temperature of Bi–sulfosalts can be considered in the range of 300 to 350°C .

5.3. Silver and gold mineralization, the supergene oxidation, and secondary enrichment of Ag and Au

Sulfide and sulfosalt mineralization at the Chodarchay deposit comprises different mineral assemblages, some of which are Ag hosts.

Pyrite is not a significant host for Ag at Chodarchay. Arsenic plays an outstanding role in the facility of trace element incorporation in the pyrite structure (Reich et al., 2005; Manuel et al., 2018); low arsenic concentrations in the pyrite samples from Chodarchay (Table 1; Fig. 8a) led to relatively clean pyrite and very few incorporated elements.

Chalcopyrite hosts Ag as a solid-solution. It as well as contains complex Ag-rich Bi-sulfosalt exsolutions. Silver is capable of partitioning into the chalcopyrite structure via simple substitution of monovalent Ag^+ to Cu^+ ion across $\text{Ag}^+ \leftrightarrow \text{Cu}^+$ (Li et al., 2013; Nelson, 2013; Reich et al., 2013). For the dominant partitioning of Ag into sulfides, the precipitation temperature should be higher than 200°C (Guangzhi, 1996). Fluid temperature for the Ag and chalcopyrite precipitation is in the range of 200 to 350°C ; on the higher head of this temperature limit, trace element contents occur as a solid-solution within the chalcopyrite and usually decrease with temperature dropping (Simon et al., 2000; George et al., 2018; Sykora et al., 2018). When the hydrothermal fluid cools down, and the chalcopyrite solidifies, trace elements start to appear, distinct micro-inclusions as the sulfosalt micro-inclusions. This switch from solid-solution Ag to Ag-rich micro-inclusion phase specifies a temperature reduction in the mineralizing fluid. The comparative content of Ag as a solid-solution in the chalcopyrite reduces when the relative abundance of Ag-bearing sulfosalt micro-inclusions increases. At Chodarchay, most of the Ag-bearing sulfosalt minerals formed on the margins and fractures of the chalcopyrite, indicating that the cooling began from these parts.

Epithermal deposits have higher contents of most elements as a solid-solution within the sphalerite (Cook et al., 2009), one of which can be Ag. The Chodarchay sphalerite samples have up to 0.1 wt% Ag averaging 0.01 wt%, but sphalerite at the deposit has a low abundance, so it is not mainly considered a silver host.

Mostly, Ag is recognized to enter into galena (George et al., 2015). Silver is substituted into galena, and in one spot, Ag concentration reaches up to 1.15 wt%. Galena may gather high Ag contents as a solid-solution, but Ag abundance at Chodarchay is low, and galena is also a

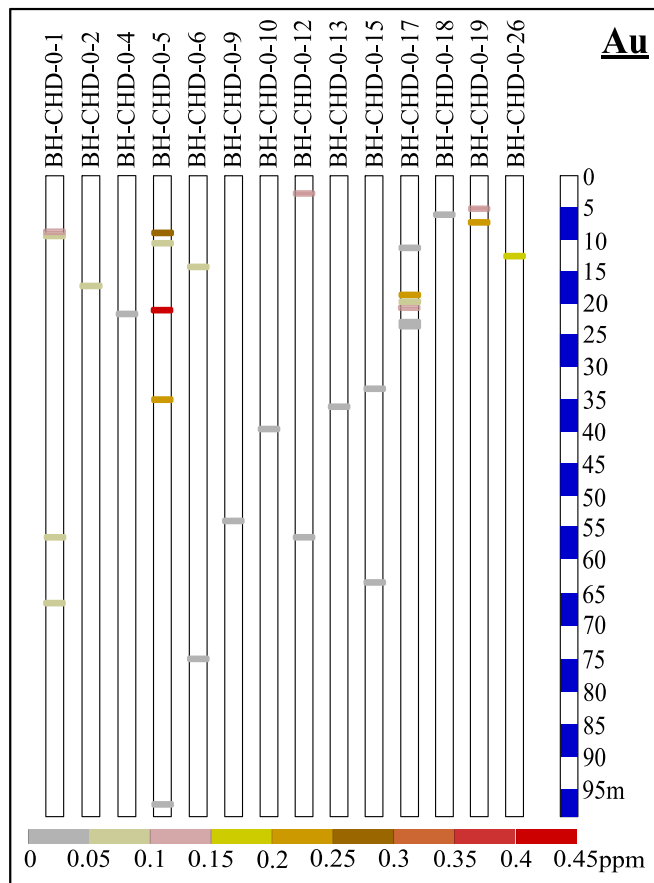


Fig. 11. Variation in the Au ($\mu\text{g/g}$) grade with increasing depth at the Chodarchay deposit.

minor phase which led to the deduction that galena is not the main host for the total Ag at the deposit. Therefore, Ag at Chodarchay has powerfully hosted in the Cu-(Fe) sulfides; most of them are chalcopyrite and Ag-rich sulfosalt micro-inclusions in chalcopyrite.

In addition to the microprobe analytical results on various sulfide and sulfosalt minerals, the results of the ICP-OES studies show that Ag demonstrates the highest correlation with Cu and Fe, respectively. Fig. 10 shows the role of chalcopyrite in hosting Ag at Chodarchay.

The highest Au content (434 ppb) in the fire-assay analyses is from sulfosalt micro-inclusion in chalcopyrite or sphalerite in sample BH-CHD-5. According to the microprobe data, Au in hypogene ore is only recorded in sphalerite, and some sulfosalts are hosted in chalcopyrite.

The supergene processes and secondary enrichment of Ag and Au can occur in the oxidation zone. The reactive Ag and Au-bearing sulfide minerals, during the supergene condition, release their geochemical budget.

Mn^{2+} in the oxidation zone can be the Ag^+ reducing agent. When Mn^{2+} and Fe^{2+} concentrations reach a specific proportion, MnO_2 and Ag precipitate in the secondary enrichment zone ($\text{Fe}^+ + \text{Ag}^+ = \text{Ag} + \text{Fe}^{3+} + \text{Mn}^{2+} + 2\text{Ag}^+ + 4(\text{OH})^- = 2\text{Ag} + \text{MnO}_2 + 2\text{H}_2\text{O}$; Palyanova et al., 2016). Supergene oxidation and secondary enrichment of Ag and MnO_2 are found in the oxidation zone of the Chodarchay deposit.

Moreover, according to the results of the fire-assay analysis of oxide samples, Au is high in secondary Fe-Mn oxide samples. However, in the supergene oxide samples that do not contain Fe-Mn, the Au contents are low. Therefore, like Ag, the precipitation of Au in the supergene oxidation zone is dependent on the presence of Mn. According to the sample depths and Au contents, the concentration of Au is greater at shallower depths (Fig. 11).

ICP-OES analyses show that compared to other base metals, Cu has

the highest correlation with Ag. These results confirm the hosting of chalcopyrite for solid-solution Ag and complex Ag-rich Bi-sulfosalt ex-solutions. Since the mineralized fluids have infiltrated most of the rock units in the region, the mineralization depends on ore-bearing brecciation and the presence of mineralized vein-veinlets in the units and not necessarily the type of rocks.

ICP-OES and Fire-assay analysis confirm that the highest correlation coefficients of elements with gold are, respectively, Mn, Cu, and Ag. These results emphasize that Au is enriched in secondary Fe-Mn oxide. Also, in hypogene mineralization, it is associated with copper and silver sulfosalts inclusions inside chalcopyrite.

5.4. Comparison of the Chodarchay sulfosalts with those from other epithermal deposits in Iran

Iran is geologically divided into different structural zones. The zones containing various subtypes of epithermal deposits in Iran are: 1) Urumieh-Dokhtar, 2) Arasbaran, 3) Alborz (Tarom, Torud–Chah Shirin, Binalood subzones), 4) Sabzevar-Taknar, 5) Sanandaj-Sirjan, 6) Makran, 7) Lut; and 8) Central Iran. The location of Iran's epithermal deposits is shown in Fig. 12. The low-, intermediate-, and high-sulfidation types of epithermal mineralization, based on the characteristics of the deposits, have been discussed in a book on the epithermal deposits of Iran (Ghaderi et al., 2023).

In the following, most of the epithermal deposits of each zone and the sulfosalt minerals reported from each deposit are mentioned:

Urumieh-Dokhtar zone (N = 16):

Low-sulfidation deposits: Chah Zard (LS-IS; Ag sulfosalts–tennantite–tetrahedrite; Kouhestani et al., 2017b), Kalchoueieh, Chah Alikhan, Narbaghi (tennantite–tetrahedrite; Fazli et al., 2019), Shojaabad, Milajerd and Shahrzad;

Intermediate-sulfidation deposits: Goumash Tappeh (tetrahedrite; Salehi et al., 2010), Kourcheshmeh (tennantite–tetrahedrite; Khanmahdloo et al., 2023), Lak, Nadooshan, and Qoyjeh Yeylaq;

High-sulfidation deposits: Atash-Anbar (HS-IS; tetrahedrite–freibergite–friedrichite; Tale Fazel et al., 2023), Chah Mesi (tennantite–tetrahedrite, enargite–luzonite–famatinite–Ag-rich sulfosalts; Altenberger et al., 2022), Chahar Gonbad, and Darreh Zar;

In this zone, the presence of sulfosalt minerals has been reported from 6 deposits of three subtypes, which are: Ag sulfosalts–tennantite–tetrahedrite–freibergite–friedrichite–enargite–luzonite–famatinite.

Arasbaran zone (N = 14):

Low-sulfidation deposits: Siahkalan, Mirkuh Ali Mirza, Zaglik, Safi-khanloo, Esmail Kandi, and Sharafabad;

Intermediate-sulfidation deposit: Yaraloojeh;

High-sulfidation deposits: Haj Alibeig Kandi, Mazra'eh Shadi, Agh-Daragh (tetrahedrite; Asgharzadeh Asl et al., 2017), Siah Roud, Sheikhdarabad (tetrahedrite; Nokhodchi, 2004), Masjed Daghi and Miveh Roud;

In this zone, the presence of sulfosalt minerals has been reported from 2 deposits of a high-sulfidation subtype, which is: tetrahedrite.

Alborz zone (N = 25):

a) Tarom subzone (N = 14):

Low-sulfidation deposits: Khalyfehlou, Chargar (watanabeite – wittenite; Mousavi Motlagh and Ghaderi, 2019), Lubin-Zardeg (tennantite–tetrahedrite; Zamanian et al., 2019) and Nikoueieh.

Intermediate-sulfidation deposits: Changoreh (tennantite–tetrahedrite; Siahcheshm et al., 2018), Zajkan, Abbasabad (tetrahedrite–tennantite; Jahandideh Kazempour et al., 2011), Varmazyar, Shah-Ali-Beiglou (tetrahedrite–tennantite–enargite–seligmannite; Mikaeili et al., 2018), Barik-Ab (tetrahedrite; Ajayebi, 2011), Zehabad (tennantite–tetrahedrite–sulfosalt; Shabbazi et al., 2019) and Marshoun;

High-sulfidation deposits: Goloujeh (Glojeh) (Mehrabi et al., 2016), Jizvan (tetrahedrite–tennantite; Golshani Nasab et al., 2023),

In this subzone, the presence of sulfosalt minerals has been reported

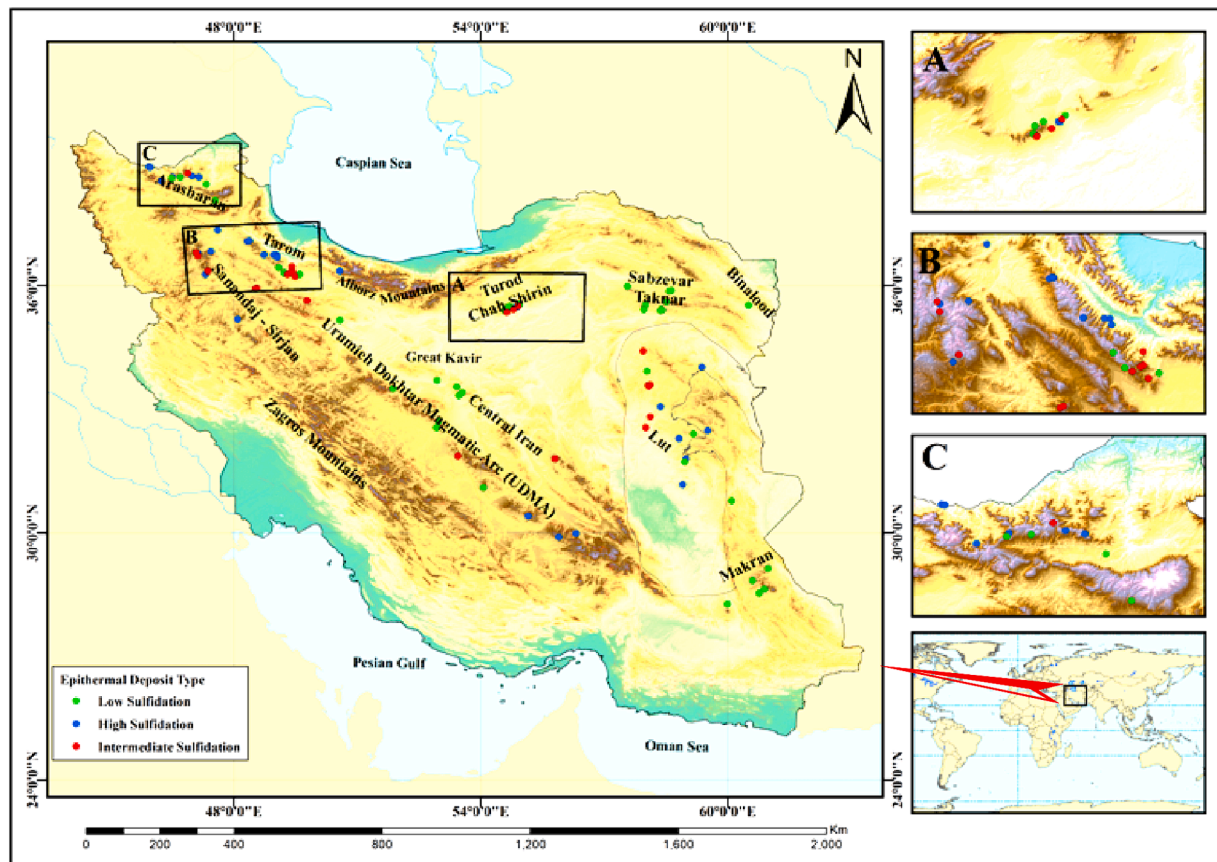


Fig. 12. The location of Iran on the global map and Iran's known epithermal deposits on Iran's geological map (modified after Vernant et al., 2004); Green, red, and yellow signs indicate low-, intermediate-, and high-sulfidation epithermal deposits, respectively. In places where deposits are dense and close together, they are shown in the appendix with a more appropriate scale (A, B, and C).

from 8 deposits of three subtypes, which are: watanabeite-wittichenite-tennantite-tetrahedrite enargite-seligmannite and an unknown sulfosalt.

b) Torud–Chah Shirin (N = 11):

Low-sulfidation deposits: Challu (tetrahedrite; Mehrabi et al., 2014), Darestan-Baghoo, Parsian, Saghari, and Abghareh;

Intermediate-sulfidation deposits: Ghole Kaftaran, Cheshmeh Hafez (tennantite–tetrahedrite; Mehrabi et al., 2014), Abolhassani and Gandy;

High-sulfidation deposits: Chahmoosa and Kalateh Chahmoosa; In this subzone, the presence of sulfosalt minerals has been reported from 2 deposits of low- and intermediate-sulfidation subtypes, which are: tennantite–tetrahedrite.

c) Binalood (N = 1):

Low-sulfidation deposit: Alamjogh.

Sulfosalt minerals have not been reported in this subzone.

Sabzevar-Taknar zone (N = 10):

Low-sulfidation deposits: Klatch Taymor, Cheshmeh Nogreh (tetrahedrite, Jamsonite; Lotfi, 2018), Arghash, Sebandoon, Bardaskan (tetrahedrite; Hashemi et al., 2010), Hassan-abad, Halak-abad, Chalpo-Kalateh Choobak, Bagh Ghazi and Daman Ghor;

In this subzone, the presence of sulfosalt minerals has been reported from 3 deposits of the low-sulfidation subtype, which are tetrahedrite and Jamsonite.

Sanandaj-Sirjan zone (N = 6):

Intermediate-sulfidation deposits: Daghdali, Ay-Qalasi (tennantite–tetrahedrite; Mohammadi Niaei et al., 2015) and Arpa Chay;

High-sulfidation deposits: Agh Otagh (tennantite–tetrahedrite; Rahmati et al., 2017), Touzlar (tennantite–tetrahedrite-enargite; Heidari et al., 2015) and Arabshah (sulfosalts; Heidari et al., 2017);

In this zone, the presence of sulfosalt minerals has been reported

from 4 deposits of intermediate and high-sulfidation subtypes, which are tennantite–tetrahedrite-enargite-unknown sulfosalts.

Makran zone (N = 7):

Low-sulfidation deposits: Bazman North, Chahnaly, Sefid-sang–Dargiaban, Shoor Chah, Kharestan, Siah Jangal-Sar Kahno, and Bidstar;

Sulfosalt minerals have not been reported from this zone.

Lut zone (N = 15):

Low-sulfidation deposits: Koodakan, Khoonik (tetrahedrite; Samiee et al., 2019), Sefidabeh and Choupan;

Intermediate-sulfidation deposits: Neian (tennantite–tetrahedrite; Hadizadeh et al., 2015), Hoze Raeis, Sechangi, Gale Chah (tetrahedrite; Mehrabi et al., 2011) and Shurab (tennantite–tetrahedrite; Mehrabi et al., 2011);

High-sulfidation deposits: Koli, Qaleh-Zari (Tetrahedrite-aikinite, matildite; Hassan Nezhad and Moore, 2006), Sar Chah, Mahour, Shekaste Sabz, and Kouhe Shah;

In this zone, the presence of sulfosalt minerals has been reported from four deposits of low- and intermediate-sulfidation subtypes, which are: tennantite–tetrahedrite, aikinite, and matildite.

Central Iran zone (N = 4):

Low-sulfidation deposits: Talkkeh, Kan Mes, and Gowd-e-Morad;

Intermediate-sulfidation deposit: Kharanghoun;

Sulfosalt minerals have not been reported from this zone.

Sulfosalt minerals from the epithermal deposits of Iran are mainly tennantite–tetrahedrite. Watanabeite and wittichenite were only identified at the Chargar deposit (Tarom subzone). Seligmannite was only reported at the Shah-Ali-Beiglou deposit (Tarom subzone). Freibergite-friedrichite were reported at the Atash-Anbar deposit (Urumieh-Dokhtar zone). Luzonite-famatinitite exist in Chah Mesi deposit (Urumieh-Dokhtar

Table 5
Sulfosalt features of most of the epithermal deposits of Iran.

Deposit name	Commodity	Sulfosalt Minerals	References
Urumieh-Dokhtar zone			
Chah Zard	Ag-Au	Tennantite, Tetrahedrite	Kouhestani et al., 2017b
Narbaghi	Cu	Ag-enriched Tetrahedrite, Tennantite	Fazli et al., 2019
Gumosh Tapeh	Zn-Pb	Tetrahedrite	Salehi et al., 2010
Kourcheshmeh	Pb-Zn	Tennantite–Tetrahedrite	Khanahmadlou et al., 2023
Atash-Anbar	Au	Tetrahedrite- Freibergite- Friedrichite	Tale Fazel et al., 2023
Chah Mesi	Cu	Tetrahedrite, Tennantite, Enargite- Luzonite- Famatinitite-Ag-rich sulosalts	Altenberger et al., 2022
Arasbaran zone			
Agh-Daragh	Cu-Au	Tetrahedrite	Asgharzadeh Asl et al., 2017
Sheikhdar Abad	Cu	Tetrahedrite	Nokhodchi, 2004
Alborz zone			
Tarom subzone			
Chargar	Au-Cu	Sulfoarsenids, Betekhtinite, Watanabeite, Wittichenite, Lautite	Mousavi Motlagh and Ghaderi, 2019
Lubin-Zardeg	Cu-Au	Tennantite, Tetrahedrite	Zamanian et al., 2019
Changoreh	Pb-Zn	Tetrahedrite, Tennantite	Siahcheshm et al., 2018
Abbasabad	Pb-Zn	Tetrahedrite, Tennantite	Jahandideh Kazempour et al., 2011
Shah-Ali-Beiglou	Zn-Pb	Tetrahedrite–Tennantite-Enargite- Seligmannite	Mikaeili et al., 2018
Barik-Ab	Pb-Zn	Tetrahedrite	Ajayebi, 2011
Zeh Abad	Pb-Zn	Tennantite, Tetrahedrite, Sulfosalt	Shahbazi et al., 2019
Jizvan	Cu	Tetrahedrite, Tennantite	Golshani Nasab et al., 2023
Chodarchay	Cu	Cuprobismuthite, Hodrushite, Paderaitite, Aikinite, Friedrichite, Emplectite, Makovickyite, Galenobismutite, Gustavite, Heyrovskyite, Neyite, Cupropavanite, Watanabeite, Colusite	This study
Torud–Chah Shirin subzone			
Challu	Cu-Au	Tetrahedrite	Mehrabi et al., 2014
Cheshmeh Hafez	Pb-Zn	Tetrahedrite, Tennantite	Mehrabi et al., 2014
Sabzevar-Taknar zone			
Cheshmeh Noghereh	Au-Cu	Tetrahedrite, Jamsonite	Lotfi, 2018
Bardaskan	Au	Tetrahedrite	Hashemi et al., 2010
Sanandaj-Sirjan zone			
Ay Qalasi	Pb-Zn	Tennantite, Tetrahedrite	Mohammadi Niaei et al., 2015
Agh Otagh	Cu	Tetrahedrite, Tennantite	Rahmati et al., 2017
Touzar	Au	Enargite, Tetrahedrite, Tennantite	Heidari et al., 2015
Arab Shah	Au-As-Sb	sulfosalts	Heidari et al., 2017
Lut zone			
Khoonik	Au	Tetrahedrite	Samiee et al., 2019
Neian	Pb-Zn	Tetrahedrite, Tennantite	Hadizadeh et al., 2015
Gale Chah	Pb-Zn	Bourmonite, Tetrahedrite	Mehrabi et al., 2011
Shurab	Pb-Zn	Bourmonite, Tetrahedrite, Tennantite	Mehrabi et al., 2011
Qaleh-Zari	Cu	Aikinite, Matildite, Tetrahedrite	Hassan Nezhad and Moore, 2006

zone). Silver sulfosalts (in general) have also been reported from the Chah Zard and Chah Mesi deposits (Urumieh-Dokhtar zone). Jamsonite occurred only at the Cheshmeh Noghereh deposit (Sabzevar-Taknar zone). Aikinite and matildite were observed only at the Qaleh-Zari deposit (Lut). Enargite occurred at the Touzar (Sanandaj-Sirjan zone), Chah Mesi (Urumieh-Dokhtar zone), Shah-Ali-Beiglou, and Barik-Ab deposits (Tarom subzone). Fourteen types of sulfosalts have been detected by EPMA at Chodarchay (Table 5). One of the reasons for this diversity can be the overprinting of epithermal on porphyry mineralization in the region. Also, in most studies conducted on other epithermal deposits in Iran, the investigation of sulfosalt minerals has not been prioritized, and it has often been accomplished using optical microscope studies. Therefore, specialized studies on sulfosalt minerals will increase their number and diversity in other epithermal deposits.

6. Conclusions

The EPMA analytical results for samples from the Chodarchay deposit supply considerable knowledge concerning the mineralization states and chemical changes in the minerals. The deposit is not only described by usual base metal sulfides such as sphalerite, galena, and chalcopryrite but also contains sulfosalt accessory minerals that host various trace elements. Among more than 200 mineral species (Moëlo et al., 2008) in the sulfosalt group members, Bi-rich assemblages are the most common at Chodarchay. The association of the twelve Bi-sulfosalts was identified, and an interesting assemblage of sulfosalts was found. Bismuth minerals at Chodarchay are classified into four types: 1) Cuprobismuthite series; 2) Cu-Bi-S such as emplectite; 3) Pb-Bi-S such as neyite; and 4) Aikinite–bismuthinite series (aikinite and friedrichite). Cuprobismuthite series are the most common and occur in paragenesis with other sulfosalts and chalcopryrite. Their occurrence is characterized by a high Cu content and the predominance of chalcopryrite over other base metal sulfides. Watanabeite and colusite sulfosalts also occurred at Chodarchay. The occurrence of Bi-sulfosalts can propose temperatures of ~ 350 °C for the deeper parts of the deposit.

Measurable trace concentrations of Au are noted in the sphalerite, Bi-Pb-(Cu), Cu-Bi sulfosalts, and some oxide minerals, implying that these minerals may be minor Au hosts. Silver is partitioned into most of the ore minerals but preferentially into chalcopryrite as a solid-solution, and Ag-bearing sulfosalts occur within the chalcopryrite, and these are thus the major hosts for Ag. A high Ag/Au ratio in the ore-forming fluid and evidence for cooling-driven precipitation may be significant reasons for forming no more Au throughout the deposit. The As-poor pyrite is an Ag-poor container. Based on Ni contents in the pyrite, Ag and other ore-forming substances were principally taken from mantle-related origins. Therefore, different stages of mineralization can be observed in the deposit.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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